Can memistor-class electrochemical weights train neural networks?

A pre-registered simulation study from Widrow's 1960 memistor to modern ECRAM, with an exploratory device sensitivity analysis and an experimental roadmap

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Abstract

Widrow and Hoff stored neural-network weights in memistors, three-terminal electrochemical cells, in 1960; lithium-ion, metal-oxide and protonic ECRAM are their modern descendants. We ask, in simulation, how close such devices come to digital training when every weight update is made of device pulses and the reads are kept ideal (fp32). We compare plain SGD and three Tiki-Taka-family algorithms on five device specifications, on MNIST, a CIFAR-10 subset and a byte-level language model, with five seeds that share initialisation and data order across arms. Against SGD with exact updates at the same learning rate, 4 pairings (Li-ECRAM and protonic-inspired, each with chopped TTv2 or AGAD) stay within the pre-set margins on all three tasks, while plain SGD exceeds them on every published device. Against a separately tuned digital Adam no pairing does: on the language model the best analog result is 23.36 % worse in bits per byte. Over a factor of four in width no arm's gap grows detectably beyond the margin; endpoint equivalence was established for one CIFAR pairing only, and growth is detected for another without resolving its size against the margin. An exploratory sensitivity analysis with three seeds establishes only a few one-at-a-time ranges and shows that an earlier asymmetry limit was mostly lost weight range. Against explicit pulse streams the aggregate update meets preset point-estimate lines on 10 of 12 probes (bootstrap: 1), not where clipping removes nearly all of an update, and ignores shared-line dependence; in training one label and two close rankings change. A conditional energy projection, borrowing an analog inference chip's read efficiency, gives 17.9–32.4× less energy per weight-row than an H100, for cells that pass the margin under ideal reads. This version repairs the simulator after independent reviews and reports how far each repair moved the original results.

1 Introduction

Training a neural network on a digital processor moves every weight from memory to arithmetic and back at every update. In 1960, Widrow and Hoff took a different route: they stored each weight of their adaptive ADALINE neuron in a *memistor*, an electrochemical cell whose conductance is set by the time integral of a current through a third terminal [1–3]. The weight was read without being disturbed, and it was adapted in place. The device was macroscopic, the cells were wet, and no multilayer learning rule existed yet, so the line ended with the Madaline hardware of the early 1960s [3]. That machine was a working analog learning system.

Sixty years later, three-terminal electrochemical devices have returned at the nanoscale. Lithium-ion ECRAM [4, 5], CMOS-compatible metal-oxide ECRAM [6], protonic programmable resistors [7, 8] and open-loop programmable arrays [9] all decouple the read path from the write

path, just as the memistor did. They report hundreds to thousands of conductance levels, nanosecond pulses and femtojoule-scale write energies. In parallel, algorithms for training on imperfect analog weights have matured: the stochastic pulse-coincidence update of the resistive processing unit (RPU) [10], Tiki-Taka [11] and its successors TTv2 [12], chopped TTv2 and AGAD [13], and residual learning with convergence guarantees [14].

That electrochemical weights can train small networks is already established: Chen et al. trained in situ on an ECRAM array and simulated larger networks [9], and Rasch et al. studied reference correction, retention and device requirements for the TT family [13]. Most recently, Wu et al. [15] simulated ECRAM-calibrated Transformers of up to about 123 million parameters with finite converters and device nonidealities, keeping the weight-gradient path digital. Byun et al. [16] derived device specifications for TTv1 and TTv2 by simulation; relative to that analysis, this study covers the chopped algorithms c-TTv2 and AGAD and a catalog of published electrochemical devices, and reports seed-paired intervals against absolute margins, a language model beside the image tasks and a test of the aggregate update against explicit pulse streams; its own sensitivity analysis is narrower. Our contribution is narrower and specific: *a reproducible, seed-matched simulation comparison* of plain SGD and three device-tolerant algorithms on a catalog of published (or, where nothing is published, labelled hypothetical) electrochemical devices, with the weight update done entirely in the devices, on three small tasks; a sensitivity analysis against absolute accuracy lines; and a *conditional* energy projection. Results are stated against two digital baselines: SGD with exact updates at the same learning rate, and a separately tuned Adam.

What this paper provides:

1. **A simulator with stated limits.** Reads are ideal (fp32); only the update is analog (Fig. 1). It reproduces selected published outcomes of Gokmen and Haensch 2020 under this simplified read model, and the offset-robustness ordering of Rasch et al. 2024 (Section 4.1). After an independent review it was repaired (separate random streams, a reset made of pulses, the AIHWKIT device presets mapped one to one) and each repair is tested (Section 3, Appendix F).

2. **A seed-matched comparison** of SGD, TTv2, chopped TTv2 and AGAD on five device specifications, three tasks (MNIST, a CIFAR-10 subset, a byte-level language model on human-written text) and five seeds, with every arm of a seed sharing its initial weights and data order (Section 4.2).

3. **A width study** with per-arm learning rates and an equivalence test (Section 4.4).

4. **An exploratory device sensitivity analysis** against absolute accuracy lines, one property at a time with three seeds (not fabrication tolerances), with the asymmetry sweep repeated at a fixed symmetry point so that asymmetry is not confounded with lost weight range (Section 4.5).

5. **A conditional energy projection** for the weight-matrix operations, reported only where the device reaches the stated quality (Section 5).

6. **A proposed roadmap** of device experiments and a proposed, unbuilt bench replication protocol of the 1960 memistor (Section 7, Appendix A).

Every experiment was planned in a committed plan before it ran; the plans are internal, timestamped by the repository, not an external registry. The original runs and the revision runs are kept apart, and every deviation is recorded (Section 9). All results are simulations, and every device parameter carries a provenance label (measured, fitted, claimed, estimated or assumed).

2 Background and related work

2.1 The memistor (1960)

Widrow and Hoff introduced the ADALINE and the least-mean-squares (LMS) rule in 1960 [2]. To store its weights they built the memistor [1]. A pencil-lead (graphite) substrate sits in a copper-sulphate plating bath. Copper is plated onto it or removed from it by a current through a third electrode, and the conductance between the lead's two ends is sensed with a small AC signal that does not disturb the state. Widrow, Pierce and Angell described elements with "thousands" of analog levels that spanned roughly 100 to 1 ohm in about ten seconds at a few milliamperes; this description is reported in secondary sources, see [17]. Mouttet analyses the memistor as a three-terminal memristive system and summarises Widrow's ideal: a conductance linear in the plating charge, reversible, non-volatile and free of hysteresis [17]. Widrow and Lehr recount that the Madaline I machine was built in hardware with memistor weights and used for pattern recognition research [3]. Write noise, up/down asymmetry, retention and endurance were never reported in a form we could find, so in our catalog these are estimates (Section 3.4). The memistor predates Chua's memristor [18] and its later physical identification in titanium-oxide films [19]. It is the conceptual ancestor of today's three-terminal synaptic transistors.

2.2 Device requirements for training

Gokmen and Vlasov [10] defined the resistive processing unit (RPU). Updates are stochastic pulse trains whose coincidences change the conductance by a fixed step (stochastic computing, [20]). They derived device specifications for plain SGD on MNIST at a small error penalty: about a thousand levels, up/down asymmetry of a few percent, and tolerance to large cycle-to-cycle and device-to-device variation. Agarwal et al. [21] reached similar conclusions for conventional (SGD) training, and the requirements were extended to convolutional networks [22]. Byun et al. [16] derived such specifications for TTv1 and TTv2 by multiparameter simulation of asymmetry, retention, pulse count and cycle-to-cycle variation: in their simulations TTv2 permits more lenient specifications than TTv1, and plain SGD needs the most stringent. Reviews cover the device landscape [23–26]. Mixed-precision schemes [27] accumulate the gradient digitally and program the devices rarely. They relax the device demands but give up the fully parallel in-memory update.

2.3 Algorithms that tolerate asymmetric devices

Real devices change their conductance by different amounts up and down, and the step depends on the state (soft bounds). Under plain SGD this acts as an implicit penalty that pulls weights toward each device's symmetry point [11, 14].

- **Tiki-Taka (TTv1)** [11] uses two arrays. The gradient is accumulated on an array A, and its columns are periodically transferred into the weight array C, with $W = \gamma A + C$. Asymmetry then acts as a useful decay on A instead of a bias on W.
- **TTv2** [12] inserts a digital buffer H between the two arrays. H low-pass filters the reads of A, and a pulse is sent to C only when a threshold is crossed. TTv2 still needs every device's symmetry point to be subtracted by a reference.
- **Chopped TTv2 (c-TTv2)** and **AGAD** [13, 28] remove this sensitivity. A random sign modulates the inputs and the reads (c-TTv2). AGAD additionally replaces the static reference with a digital, on-the-fly reference taken from the recent history of A.

- **Residual learning** [14] gives convergence guarantees for this family.

IBM's open-source AIHWKIT [29] implements these algorithms with fitted device presets, including the Li-ECRAM and metal-oxide ECRAM presets whose parameters we use. In the revision these two presets are mapped one to one onto AIHWKIT's own LinearStepDevice model, with its shifted bounds and slope spread, and checked against AIHWKIT's pulse responses (Section 3.4).

2.4 Electrochemical three-terminal devices

- **Lithium-ion synaptic transistors and ECRAM.** Fuller et al. [5] showed a lithium-ion synaptic transistor, followed by an ionic floating-gate array [30]. Tang et al. [4] reported Li-WO_3 ECRAM with about a thousand symmetric, linear states and nanosecond pulses.
- **Metal-oxide ECRAM.** Kim et al. [6] made it CMOS-compatible.
- **Protonic resistors.** Onen et al. [7] reported nanosecond protonic programmable resistors with a phosphosilicate-glass electrolyte. In 2026, Xu et al. [8] quantified cycle-to-cycle (3%) and device-to-device (21%) variability of BEOL-compatible protonic ECRAM.
- **Open-loop programmable arrays.** Chen et al. [9] demonstrated a 10×10 W/YSZ/WO_x array with 256 states.
- **Organic devices.** Organic electrochemical synapses [31] extend the family to low voltages.

In-situ training on ECRAM hardware has so far been shown on small arrays, for example a 3×3 array by Li et al. [32] and the 10×10 array of [9]. Other memory technologies have gone further. Phase-change and resistive arrays have trained or run networks at larger scale [33–36], and analog-AI inference chips with on-chip periphery now report measured efficiencies of about 10 TOPS/W [37, 38].

Scaling analog training. Wu et al. [15] co-designed the weight mapping, the optimizer and the converters, and simulated ECRAM-calibrated Transformers up to about 123 million parameters with finite converter resolution and device nonidealities. Their weight-gradient path stays digital, so they do not test the fully in-memory update studied here. The two studies answer complementary questions: what an analog update saves in computation and data movement, and what it costs in accuracy and calibration. Our models are far smaller (Section 6).

3 Methods

3.1 Device model

Each analog weight is one device with a normalised state $u \in [-1, 1]$ between its own bounds. Its conductance is $g = g_{\min} + \frac{u+1}{2}(g_{\max} - g_{\min})$, and the weight is read differentially as $w = \alpha\,(g - g_{\mathrm{ref}})$. One nominal pulse moves u by $k = 2/n_{\mathrm{states}}$ at mid-range. With asymmetry a the up and down steps are $k_{\pm} = k\,s\,(1 \pm a/2)$, where s is a per-device step factor with spread σ_{d2d}. With saturation (soft bounds) the steps depend on the state:

$$\Delta u_{+} = k_{+}\,(1 - \gamma_{+} u), \qquad \Delta u_{-} = -k_{-}\,(1 + \gamma_{-} u). \tag{1}$$

The per-device slopes $\gamma_{\pm}$ have a relative spread σ_{γ}. A device's symmetry point, where an up and a down pulse cancel, is $u^{*} = (k_{+} - k_{-})/(k_{+}\gamma_{+} + k_{-}\gamma_{-})$. The reference g_{ref} is set at $g(u^{*})$ plus a

calibration error of standard deviation σ_r, in units of the half-range. A constant-step device has no symmetry point and is referenced at mid-range.

- **Several pulses in one update.** When a device receives n_+ up and n_- down pulses in one update, the pulses are taken as uniformly interleaved, and $\mathrm{d}u/\mathrm{d}s = A - Bu$ is solved exactly over the batch, with $A = k_+ n_+ - k_- n_-$ and $B = k_+ \gamma_+ n_+ + k_- \gamma_- n_-$.
- **Write noise.** Cycle-to-cycle noise adds a Gaussian with standard deviation c_{2c} times the local step, growing as the square root of the pulse count.
- **Drift.** Retention loss relaxes u toward the bare state $u = -1$ with a time constant τ in training steps.
- **Bounds.** The bounds $g_{\min}$, $g_{\max}$ carry a device-to-device spread.

The AIHWKIT LinearStepDevice (revision). The two devices taken from AIHWKIT presets (Li-ECRAM, MO-ECRAM) use AIHWKIT's own parameterisation instead: the state is the weight itself, with hard bounds $b_{\max} = w_{\max}(1 + \sigma_b \xi)$ and $b_{\min} = w_{\min}(1 + \sigma_b \xi)$ around shifted means ($w_{\max} = 1.1724$, $w_{\min} = -0.8276$ for Li-ECRAM, so that the symmetry point is at zero), steps $k(1 \pm \beta + \sigma_{\mathrm{d2d}} \xi)$, and slopes divided by the mean bounds, $\Delta w_+ = k_+(1 - |\gamma_+ + \sigma_{\gamma+} \xi|\, w/w_{\max})$ and $\Delta w_- = -k_-(1 + |\gamma_- + \sigma_{\gamma-} \xi|\, w/|w_{\min}|)$, with multiplicative write noise.

Forward and backward passes read the weights ideally, in fp32. Read noise and DAC/ADC quantisation are outside the scope of this study, which isolates the *update*; the energy projection (Section 5) nevertheless prices finite-precision reads, so accuracy and energy rest on different read assumptions. Biases, layer norms and the language model's position table are digital.

3.2 The update

We use the stochastic pulse-coincidence scheme of [10]. Each sample (row) offers $BL = 31$ pulse slots. Input i fires with probability $p_{x,i} \propto |x_i|$ and error j with probability $p_{d,j} \propto |d_j|$, with the scaling split equally between the two streams (update management). Weight ij therefore receives coincidences with mean $\eta\, |x_i d_j| / (\alpha k)$. Up and down counts are kept separate, so asymmetry is never cancelled digitally. Within a batch the counts of all samples are summed and drawn from a Poisson distribution with the requested mean. A mini-batch therefore only groups pulses; it does not average gradients.

This is a surrogate, not a pulse-by-pulse simulation. The mean is exact only while the largest pre-clamp slot probability is at most one, i.e. $(\eta/\alpha k) \max |x| \max |d| \le BL$; beyond that the probabilities are clipped and part of the update is lost. The repaired simulator records, per layer, the fraction of updates that clipped and the lost fraction of the requested update (Section 4.2). Discrete pulses do not commute on a state-dependent device (on a 20-level soft-bounds device one up then one down pulse from $u = 0.4$ ends at 0.314, the reverse order at 0.334, and the batch solution at 0.327), and independent per-cell Poisson draws ignore the correlations of shared row and column pulse streams. A larger variance than the binomial one is therefore not claimed to be conservative. Section 4.3 compares the surrogate with explicit pulse streams on training cases.

3.3 Algorithms

SGD. The pulses go directly into the weight devices.

TTv1 [11]. The pulses go into an array A, and the forward weight is $C + \gamma A$. Every batch, n_c columns j of A are read and added to C by a pulsed outer product, $\lambda A_{:,j}\, e_j^\top$. A is never reset.

TTv2 (local variant of [12]). Each read column is added to a digital buffer, $H_{:,j} \mathrel{+}= \lambda A_{:,j}/(\alpha k)$. C receives up to 31 pulses where $|H| \geq 1$, and H keeps the remainder. Optionally the read column of A is reset after the read. Published TTv2 uses single-pulse transfers, forgets a buffer element after it fires and does not reset A; we name ours a variant and run the published-style form beside it in the diagnostic (Appendix F).

The reset of A. The repaired *pulse reset* uses the transfer's own read: a sign-directed train of $\mathrm{round}(|A_{ij}|/\alpha k)$ pulses toward the read zero, re-read and corrected once, at most 62 pulses per weight, every pulse applied through the device dynamics (with its write noise, state-dependent step and device spread) and counted. The original simulator's *ideal reset* (Appendix F) is reported only as a labelled variant.

c-TTv2 [13]. Chopper signs $c_j \in \{\pm 1\}$ modulate the inputs of the A update ($c \odot x$) and demodulate the reads ($c_j A_{:,j}$). Each read column's chopper flips with probability ρ.

AGAD [13]. The choppers of c-TTv2, plus a digital leaky average of the reads, $p_j \leftarrow (1-\beta)p_j + \beta A_{:,j}$. The read is $z_j = c_j(A_{:,j} - p_j^{\mathrm{ref}})$, and p_j^{ref} is reset to p_j whenever c_j flips.

In the main experiments every TT-family A array uses a learning rate $1/\lambda$ times larger, so that $\lambda\times$ (the A rate) equals the SGD rate; the TTv1 configuration of the published benchmark (Section 4.1) is the exception ($\eta = 0.01$, $\lambda = 0.02$, A rate equal to η). The settings, including the choice between the pulse reset and no reset, were chosen once on MNIST validation with a mid-quality device that is not in the catalog (Section 3.6; Appendix B). We implemented the algorithms from the papers' text; we did not use the authors' code. Figure 1 shows which parts are simulated as device operations and which are digital.

3.4 Device catalog and provenance

Table 1 lists the devices as every revised run uses them. Every field carries a provenance label:

- MEASURED: reported in the paper;
- FIT: from the AIHWKIT presets fitted to the published devices [29];
- CLAIM: a qualitative statement in the paper, translated by us;
- ESTIMATE: ours;
- ASSUMED: for example, the 2% reference error given to every device.

The protonic device of [7] is mostly estimated: the paper reports a thousand up and a thousand down pulses that are "nearly linear and symmetric", but no write-noise or variability statistics, and its finite retention is replaced by no drift. We therefore call it a *protonic-inspired scenario*; a thousand applied pulses are not a demonstration of a thousand independently programmable states. The original field copy of the Li-ECRAM and MO-ECRAM rows is kept, labelled as superseded, in Table 17 (Appendix E); the revised runs use the one-to-one AIHWKIT mapping above (bounds $[-0.8276, 1.1724]$ and $[-0.8286, 1.1714]$, absolute slope spread 0.05). Widrow's memistor is given two specifications, one as he described it and one pessimistic, together with a pencil-lead cell built today with modern hobby materials (Appendix A).

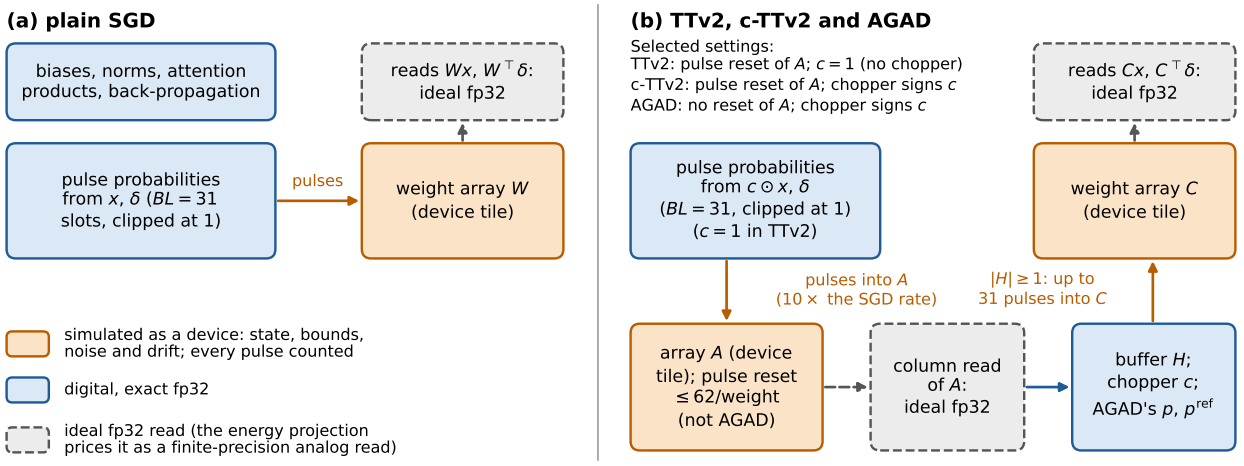


Figure 1: Device operations and digital ones, for one learned weight matrix: (a) plain SGD; (b) TTv2, c-TTv2 and AGAD as run here, with forward weight C (TTv1, used only in Section 4.1, has no buffer and no reset). In the selected configurations (Table 14) TTv2 and c-TTv2 reset A with pulses and AGAD does not reset it; in the common notation $c \odot x$, $c = \mathbf{1}$ for TTv2, and c-TTv2 and AGAD use chopper signs. Pulse probabilities are computed digitally; the pulses go through the device model; the forward and backward reads are ideal fp32 products of the device state; biases, layer norms, nonlinearities, attention products and the LM's position table are digital. The energy projection (Section 5) prices the reads as finite-precision analog reads, the pulses at device energies and, for the TT family, one ADC conversion and one buffer update per read weight.

3.5 Tasks and baselines

MNIST [39]. A 784–256–128–10 tanh MLP; 10 epochs, batch 128, on 55,000 training images. 5,000 held-out training images form the validation set, and the 10,000 test images are the test set.

CIFAR-10 subset [40]. Three 3×3 convolutions with c, $2c$ and $2c$ channels (base $c = 32$), computed as crossbar matrix products over unfolded patches, then two fully connected layers. 30 epochs, batch 128, on the first 10,000 images of training batches 1–2. The next 2,000 images are the validation set, and the 10,000-image test set is the test set.

Character LM. A byte-level, two-layer transformer [41] of width d (base 128) and context 128, trained on 8 MB of enwik8 [42], which is human-written Wikipedia text of 2006. 3,000 steps of 32×128 bytes, each window starting at a uniformly random position. The start's upper bound is one short, so the last window of the training slice is never drawn and the slice's final byte is never a target; no held-out byte is affected. Every learned weight matrix uses the device model, including the embedding and the head; the activation-dependent attention products and the other operations are digital. Validation is the slice's last 0.5 MB, held out from training in selection runs. Test is a separate 0.5 MB slice.

The first baseline ("FP-SGD") is the same network, initialisation, data order and update rule with an exact floating-point weight update. The second is Adam [43] on digital twins with the same initialisation and data order and its own learning rate selected on validation; in the revision it is reported for all three tasks, beside FP-SGD. The gap is FP accuracy minus device accuracy for

Table 1: Revised device catalog, the configuration of every revised run (simulator units; provenance per field in records/revision/CATALOG-REV.json). * AIHWKIT's presets mapped one to one onto its LinearStepDevice (shifted bounds, bound-scaled slopes, device spread 0.05 of the slopes and 0.02 of the asymmetry); the others use the paper's model with bounds $[-1, 1]$. Columns: levels n_{st}, asymmetry a (device spread), saturation γ_+/γ_-, bounds, write noise c_{2c}, step spread σ_{d2d}, reference error σ_r; no drift. The original field-copy catalog is Table 17 (Appendix E).

Device	Label	n_{st}	a	γ_+/γ_-	bounds	c_{2c}	σ_{d2d}	σ_r
Li-ECRAM*	PUBLISHED-FIT	1000	0	0.12/0.51	[-0.83, 1.17]	0.3	0.1	0.02
MO-ECRAM*	PUBLISHED-FIT	7089	0	0.42/0.73	[-0.83, 1.17]	2	0.1	0.02
Protonic-inspired	PARTLY-PUBLISHED	1000	0.05	0.20/0.20	[-1, 1]	0.3	0.1	0.02
W/WO_x ECRAM	PUBLISHED-FIT	256	0	0.63/0.63	[-1, 1]	2.048	0.023	0.02
Memistor (optimistic)	CLAIM+ESTIMATE	1000	0.05	0.10/0.10	[-1, 1]	0.3	0.2	0.02

classification, in percentage points. For the LM it is the device's bits per byte relative to FP's, minus one, in percent.

3.6 Protocol and statistics

Every experiment was pre-registered in a committed plan before it ran (Section 9).

- **Hyperparameters** were chosen on validation splits only, with an edge rule: whenever the best value of a grid lies at its edge, the next half-decade point is added. The learning rate of every device arm in the main table is the FP validation choice. The scale study instead tunes each arm separately.
- **Statistics.** The main table uses five seeds per arm. Gaps are paired by seed, and we report the mean and a 95% Student-t interval (the t quantile computed exactly). A cell PASSES when the whole interval lies below the gap line (2 pp for classification, 5% for the LM), FAILS when it lies wholly above, and is UNRESOLVED otherwise.
- **Multiplicity.** We make no correction for multiplicity. The intervals describe each cell, not the family, and a best cell chosen among many is optimistic.
- **Random streams (revision).** Each seed defines separate generators for the logical initialisation, the data order, each tile's device population and the pulse and noise draws, so the paired gap compares runs that share their initial weights and their data.
- **Width (revision).** The endpoint contrast D (gap at the widest minus the narrowest width) is tested for equivalence (TOST at 5 %, margins equal to the gap lines); "no evidence of growth beyond the margin" when only the upper bound is under the margin; never "flat".
- **Sensitivity (revision).** Each grid point is labelled against the *absolute* line with 3 seeds paired with FP-SGD of the same seed; a threshold is the weakest value that passes with every value toward the best end passing.

3.7 Simulator checks

The simulator is checked by 14 checks, and each check is shown to fail on at least one deliberately planted fault (17 faults in all). Examples of what the checks cover:

- the exact-update arm equals PyTorch SGD;
- the mean pulse update equals the SGD step;
- alternating pulses converge to the analytic symmetry point, including with per-device slopes;
- write noise grows as $\sqrt{n}$;
- drift relaxes to the bare state;
- a static read offset is integrated by TTv2, mostly cancelled by c-TTv2 and nearly fully cancelled by AGAD;
- TTv1 moves C along $+\lambda A$;
- the SGD and TTv2 paths are bit-identical to the earlier code that produced our preliminary results.

The planted faults include a swapped pulse sign, a dropped chopper demodulation and a never-updated AGAD reference. Both suites run at the start of every compute session and stop it on any failure (Appendix C). The revision adds 8 checks of the repairs (matched initialisation and data order across arms, a shared device population across algorithms, resets that charge exactly the pulses they apply, the AIHWKIT LinearStepDevice recurrence, the clipping statistics, the published-style buffer, the re-centred symmetry point, and a bit-exact replay of the original simulator), each shown to fail on the old behaviour or a planted fault (Appendix F). These checks test the implementation; they do not validate the physical model.

4 Results

4.1 Simulator validation

Gokmen and Haensch 2020. We map their RPU-baseline device onto our model:

- step 10^{-3} at $w = 0$;
- mean saturation slope 1.66 with a 25% per-device spread;
- 30% step spread and 30% cycle-to-cycle noise;
- a symmetry point at $w = 0$ for every device.

We train their 784–256–128–10 sigmoid network on MNIST for 50 epochs at their learning rate of 0.01. The results are in Table 2 and Fig. 2a.

- FP reaches 1.78% test error, against their 2.0%.
- SGD on the device reaches 14.47% [12.68, 16.26], against "about 15%".
- Tiki-Taka reaches 2.26% [2.13, 2.39], against "close to 2%".

By our pre-registered criteria the agreement is **quantitative**: this is agreement with selected published outcomes under a simplified read model, not a full quantitative validation. Matching an endpoint under different sources of error cannot rule out compensating differences, and the batch-64 result below shows how sensitive the SGD arm is to the protocol. These validation runs used the original simulator (TTv1 has no reset, so the reset repair does not touch them). The known differences are listed in Section 6.1:

Table 2: Validation against Gokmen and Haensch 2020 (FCN 784-256-128-10, MNIST, 50 epochs): test error in %, our mean and 95% interval over 3 seeds at batch 16, and one seed at batch 64.

Arm	Published [11]	Ours (batch 16)	Ours (batch 64)
FP (exact update)	2.0	1.78 [1.75, 1.81]	1.79 (n=1)
SGD on the RPU-baseline device	about 15	14.47 [12.68, 16.26]	24.32 (n=1)
Tiki-Taka (TTv1)	close to 2	2.26 [2.13, 2.39]	2.07 (n=1)
TTv2 (ours, 16 columns per batch)	–	2.59 [2.42, 2.75]	not run

Table 3: Qualitative check of Rasch et al. 2024 (Fig. 5): a 20-state soft-bounds device with an uncorrected reference offset σ_r; test error in % (mean and 95 % t interval, $n = 2$; 10 epochs). FP: 2.69 [0.47, 4.92].

Offset	TTv2	c-TTv2	AGAD
$\sigma_r = 0$	4.12 [2.79, 5.46]	4.29 [3.91, 4.67]	8.47 [7.58, 9.36]
$\sigma_r = 0.5$	24.27 [18.23, 30.30]	13.02 [10.99, 15.05]	7.88 [3.49, 12.26]

- we use mini-batches of 16, not 1;
- we do not model read noise, ADC/DAC quantisation or analog biases.

At batch 64 the FP and TTv1 results barely move, but SGD on the device degrades further. Summing more samples' pulses before the weight changes hurts a strongly asymmetric device, which is another reason to use batch 16 as the closer match to their batch 1.

Rasch et al. 2024, qualitatively. We use a 20-state soft-bounds device with an uncorrected reference offset of $\sigma_r = 0.5$ (Table 3, Fig. 2b).

- TTv2's test error rises from 4.12% to 24.27%.
- c-TTv2 rises less, from 4.29% to 13.02%.
- AGAD reads 8.47% versus 7.88% (no change detected with two seeds).

This is the published ordering; our pre-registered check reads **reproduced**. It rests on two seeds per condition and an ordering, so it does not establish that our c-TTv2 and AGAD are equivalent to the published implementations, and the absence of a detected change does not show that AGAD is unaffected. In our implementation AGAD starts from a higher error at zero offset on this very coarse device.

The device presets against AIHWKIT itself (revision). On a rented machine (never on our desk) we installed AIHWKIT 1.1.0 from PyPI and drove its EcRamPresetDevice and EcRamMOPresetDevice through long up, down and alternating pulse schedules (500 devices each, with and without cycle-to-cycle noise), using its own plotting utilities. Our AIHWKIT-mode tile, driven through the same schedules, matches AIHWKIT's mean responses, device spreads and reached bounds within the pre-registered tolerance for both presets (largest mean difference 0.0111 for Li-ECRAM and 0.0113 for MO-ECRAM, in weight units; the pre-registered tolerance is $\max(4\,\mathrm{SE}, 2\,\Delta w_{\min})$ at every stored pulse). The original field-copy mapping differs by up to 0.174 for Li-ECRAM, mostly at the bounds (Appendix F).

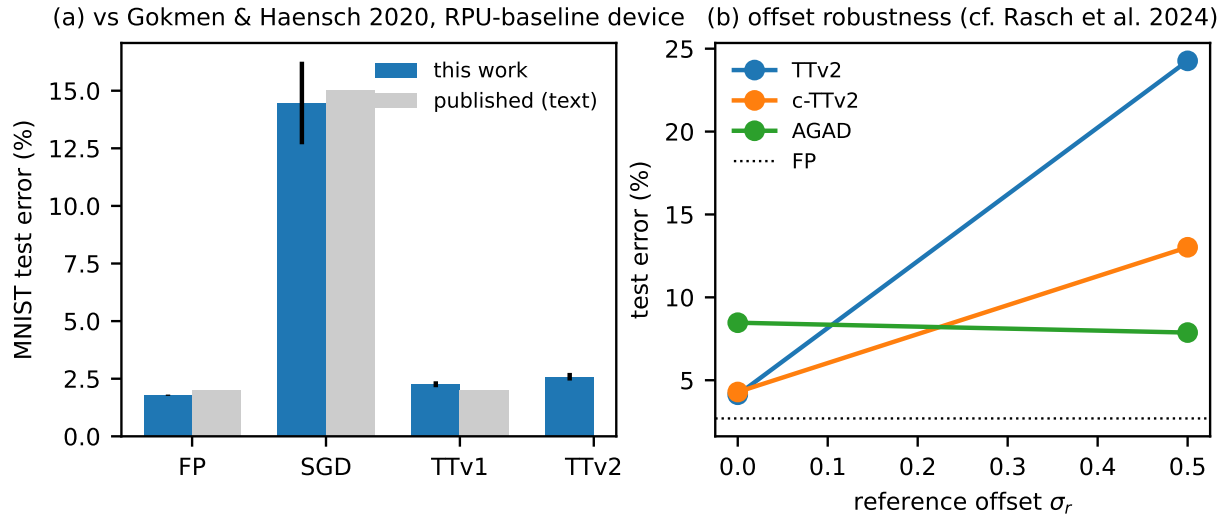


Figure 2: Simulator validation. (a) Test error of our simulator (mean and 95% interval, 3 seeds) against the values stated in the text of Gokmen and Haensch 2020 for their FCN on MNIST with the RPU-baseline device. (b) Robustness to an uncorrected reference offset on a 20-state device (2 seeds), the qualitative trend of Rasch et al. 2024, Fig. 5.

4.2 Main comparison on the repaired simulator

Tables 4 and 5 give the gap of every device–algorithm pairing over five seeds, against FP-SGD (the same network, learning rate, initialisation and data order with an exact update) and against digital Adam at its own selected learning rate. Every arm of a seed now shares its initial weights and its data order (checked by hash in every record). The FP-SGD learning rate is reused for every device arm: this is a restricted question (what happens when the FP-SGD rate is reused), not a fair-tuning comparison of the algorithms; Section 4.4 tunes each arm separately.

Against FP-SGD.

- **Plain SGD exceeded the line** in 12 of the 12 cells of the four published devices, at the FP-selected learning rate.
- **All three tasks:** 4 pairings pass on every task: Li-ECRAM c-TTv2, Li-ECRAM AGAD, Protonic-inspired c-TTv2, Protonic-inspired AGAD. No TTv2 pairing does: with the pulse reset, TTv2 fails on CIFAR on every published device.
- **Per task,** 14, 6 and 4 of the 20 device cells pass on MNIST, CIFAR and the LM. W/YSZ/WO_x fails on CIFAR and the LM with every algorithm, and MO-ECRAM fails on the LM.
- **Best cells:** Memistor (optimistic) c-TTv2 on MNIST (0.18 pp), Protonic-inspired AGAD on CIFAR (-1.34 pp) and Protonic-inspired c-TTv2 on the LM (-1.53 %). A negative gap whose interval includes zero is not evidence of superiority; and because these are the best of many cells, they are optimistic.

Against Adam. Digital Adam is the stronger baseline. On the LM FP-SGD is 25.35 % worse in bits per byte than Adam, and **no analog cell comes within the 5 % line of Adam**: the best, Protonic-inspired c-TTv2, is 23.36 % worse (0 of 20 cells pass). On MNIST and CIFAR, where Adam and FP-SGD are close, 14 and 4 cells pass against Adam. No pairing passes all three tasks against Adam (0). The analog results are therefore close to the FP-SGD comparator on these tasks, not to the best digital training in this study.

The margins. The lines are practical non-inferiority margins, not equivalence: on MNIST a 2 pp loss from 97.77 % roughly doubles the error rate. UNRESOLVED means insufficient evidence against the margin, never adequate performance. The mean digital baselines are 97.77 % and 97.75 % on MNIST, 59.69 % and 60.76 % on CIFAR, and 2.965 and 2.366 bits per byte on the LM (FP-SGD and Adam). The FP-SGD seed standard deviation is 1.02 pp on CIFAR and 0.078 bits per byte on the LM.

What the update loses. Two properties of the repaired simulator matter for reading these results.

- **Clipping.** The TT family drives its A arrays at ten times the SGD rate, which exceeds the $BL = 31$ pulse budget in the largest layers: the mean lost fraction of the requested A update reaches 57.5 % in the CIFAR output layer and 27.8 % in the LM embedding (plain SGD at most 0.7 % in any layer's mean). These are means over the device runs of one layer; the worst single layer record loses 96.4 % (MO-ECRAM AGAD, CIFAR output layer). The logged fraction is the discarded unsigned expected pulse mass, not the norm of the lost signed gradient after cancellation. Clipping the line probabilities at one is the update-management policy chosen here (pulse length and update management are configurable choices [29]), not a property of the devices; it caps the effective A rate in those layers, and the aggregate update itself is an approximation (Section 4.3).
- **Reset residual.** With at most 62 pulses per weight, the pulse reset leaves a residual on the read A columns. Averaged with every layer and run weighted equally (a macro-average of the per-layer means), it is 67.00 nominal steps; pooled over all reset events it is 26.97 (Li-ECRAM 3.7, protonic-inspired 5.5, MO-ECRAM 113.4); the largest layer mean is 2081.42 (MO-ECRAM). Reset pulses are 3.1 % of all pulses as the mean of the per-run fractions, and 0.49 % pooled (all reset pulses over all pulses). The macro-averages say what a typical layer and run see; the pooled numbers what the arrays receive in total.

Ideal-device reference. The same algorithms on an ideal device (symmetric, constant step, 1000 levels, no noise, no spread, exact reference) are the rows "Ideal 1000-level" of the tables (deviation R2). On MNIST every algorithm passes on it. On CIFAR and the LM none does: plain SGD loses 6.32 pp and 12.14 % (stochastic pulse updates on a 1000-level grid at the FP-SGD rate), and c-TTv2 and AGAD lose 18.04 and 19.20 pp on CIFAR, far more than on the saturating Li-ECRAM and protonic-inspired devices. With the settings selected here, the TT family relies on the device's own saturation as the decay of its accumulator; the ideal device has none. This is a property of these algorithm settings, not a general requirement that devices saturate: a sufficiently precise symmetric device can implement SGD, as the MNIST row shows. This reference keeps stochastic pulse counts, bounds, transfer thresholds and the selected update management, so it is not an exact-update, algorithm-only TT reference, and its MNIST success does not show that every other failure is an unavoidable device limitation.

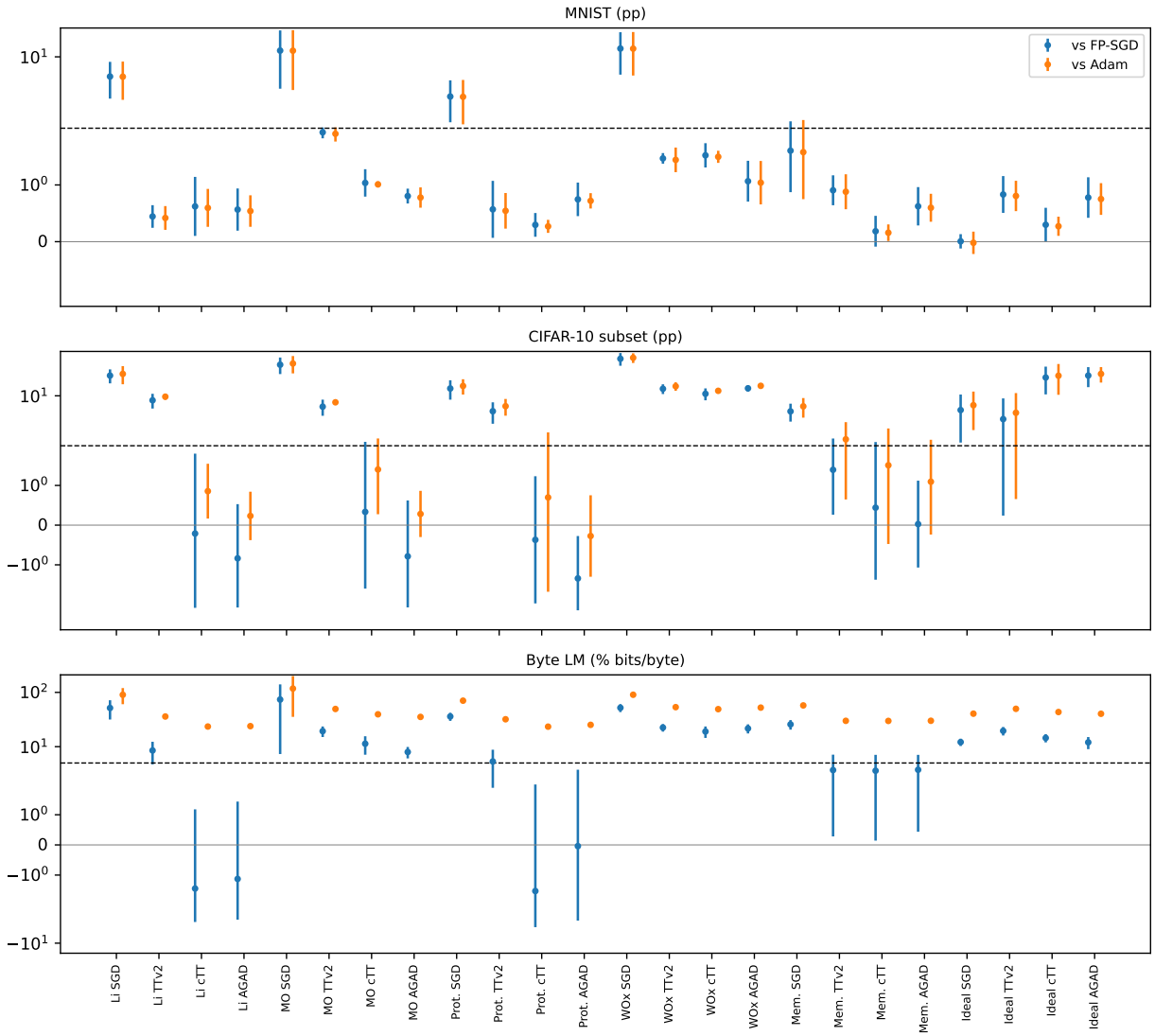


Figure 3: Repaired simulator: gap to FP-SGD (blue) and to digital Adam (orange) per device and algorithm, mean and 95 % interval over 5 seed-matched runs (symmetric-log axis); the dashed line is the gap line.

4.3 Fidelity of the aggregate update

The simulator draws each batch's pulses as independent Poisson counts per cell and applies them in one exact step of the interleaved dynamics (Section 3.1). A physical stochastic-pulse array instead fires row and column lines per slot and per sample; a cell is pulsed when both of its lines fire, and the samples follow one another. We compared the two, pre-registered with pass lines (`PLAN-REVISION-2.md`, experiment A), on the operating regime of the main table.

Update level (A1). At two points of training, one batch's inputs, errors and array state were taken from surrogate training runs and applied repeatedly from the same state by three methods: the surrogate; the explicit stream applied pulse by pulse (Bernoulli line firings, the per-pulse device

Table 4: Revised main table (repaired simulator): gap to FP-SGD at the same learning rate, mean [95 % t-interval] over 5 seeds paired by seed (same initialisation and data order); MNIST and CIFAR in accuracy points, LM in % of bits per byte; P/F/U = PASS/FAIL/UNRESOLVED against the absolute lines 2 pp, 2 pp, 5 %.

Device	Alg.	MNIST	CIFAR	LM
Li-ECRAM	SGD	6.41 [3.90, 8.93] F	19.19 [14.94, 23.44] F	51.64 [31.63, 71.64] F
Li-ECRAM	TTv2	0.44 [0.24, 0.64] P	8.65 [6.64, 10.66] F	8.51 [4.72, 12.30] U
Li-ECRAM	c-TTv2	0.62 [0.10, 1.14] P	-0.21 [-2.22, 1.80] P	-1.45 [-4.07, 1.18] P
Li-ECRAM	AGAD	0.57 [0.19, 0.94] P	-0.84 [-2.20, 0.53] P	-1.13 [-3.69, 1.44] P
MO-ECRAM	SGD	11.51 [4.88, 18.14] F	27.24 [20.15, 34.34] F	73.84 [7.32, 140.36] F
MO-ECRAM	TTv2	1.93 [1.82, 2.03] U	7.04 [5.25, 8.83] F	19.31 [15.05, 23.57] F
MO-ECRAM	c-TTv2	1.04 [0.79, 1.28] P	0.33 [-1.60, 2.27] U	11.33 [7.11, 15.55] F
MO-ECRAM	AGAD	0.80 [0.67, 0.93] P	-0.79 [-2.19, 0.62] P	7.98 [6.08, 9.87] F
Protonic-inspired	SGD	4.09 [2.29, 5.88] F	12.66 [8.83, 16.49] F	35.99 [29.87, 42.10] F
Protonic-inspired	TTv2	0.57 [0.07, 1.07] P	6.08 [4.06, 8.10] F	5.34 [1.89, 8.78] U
Protonic-inspired	c-TTv2	0.30 [0.09, 0.51] P	-0.37 [-1.97, 1.23] P	-1.53 [-5.07, 2.01] P
Protonic-inspired	AGAD	0.75 [0.45, 1.04] P	-1.34 [-2.41, -0.27] P	-0.04 [-3.84, 3.76] P
W/WO$_x$ ECRAM	SGD	12.06 [6.69, 17.42] F	33.00 [26.34, 39.67] F	52.03 [43.34, 60.71] F
W/WO$_x$ ECRAM	TTv2	1.47 [1.37, 1.56] P	12.53 [10.59, 14.48] F	22.48 [18.92, 26.04] F
W/WO$_x$ ECRAM	c-TTv2	1.52 [1.31, 1.74] P	10.67 [8.65, 12.69] F	19.02 [14.49, 23.55] F
W/WO$_x$ ECRAM	AGAD	1.07 [0.71, 1.43] P	12.72 [11.44, 14.01] F	21.64 [17.52, 25.76] F
Memistor (optimistic)	SGD	1.60 [0.87, 2.34] U	6.05 [4.35, 7.75] F	25.72 [20.59, 30.86] F
Memistor (optimistic)	TTv2	0.91 [0.64, 1.17] P	1.39 [0.26, 2.53] U	3.70 [0.28, 7.12] U
Memistor (optimistic)	c-TTv2	0.18 [-0.09, 0.46] P	0.44 [-1.37, 2.25] U	3.61 [0.14, 7.07] U
Memistor (optimistic)	AGAD	0.62 [0.29, 0.96] P	0.03 [-1.07, 1.12] P	3.75 [0.44, 7.07] U
Ideal 1000-level	SGD	0.01 [-0.12, 0.13] P	6.32 [2.22, 10.42] F	12.14 [10.28, 14.00] F
Ideal 1000-level	TTv2	0.83 [0.51, 1.16] P	4.72 [0.24, 9.20] U	19.55 [16.07, 23.03] F
Ideal 1000-level	c-TTv2	0.30 [-0.00, 0.60] P	18.04 [10.46, 25.61] F	14.47 [11.99, 16.96] F
Ideal 1000-level	AGAD	0.78 [0.42, 1.14] P	19.20 [13.21, 25.19] F	12.02 [9.00, 15.03] F
Digital	Adam	0.03 [-0.18, 0.23] P	-1.07 [-2.65, 0.52] P	-20.19 [-22.39, -17.99] P

step, write noise and bounds after every pulse; the literal stream); and a per-sample form of the same stream (seq), which replays the literal stream's line firings. Within one sample every pulse of a cell has the same sign, so seq applies the sample's train in closed form. Its write noise is an approximation: it is added once per sample at the pre-sample state (standard deviation $c_{2c}\sqrt{n}$ times the local step for n pulses), and the bounds are applied after each sample, where the literal stream applies noise and bounds after every pulse. The cases cover clipped and unclipped layers: MNIST Li-ECRAM c-TTv2 (first and output layer), CIFAR MO-ECRAM AGAD (fully connected and output layer; the worst-clipped layer of the main table) and the LM protonic-inspired c-TTv2 (embedding and output layer), 12 probes. Because seq replays the literal stream's line firings, the two are compared paired by draw, with the draws kept (16 per probe) and bootstrap bounds over whole draws (Tables 6 and 7; `PLAN-REVISION-3.md`); the draws are in the deposit inventory, and Tables 6–8 can be rebuilt from them and the probe records' metadata (Section 9); an earlier, unpaired analysis is superseded (Appendix F).

- The surrogate, whose draws are independent of the literal stream's, meets the pre-registered lines (realised gain along the requested update within 10 %, mean difference beyond sampling noise at most 0.10, median variance ratio in $[0.5, 2]$) on 10 of 12 probes, including layers that lose up to 37.5 % of the requested pulse mass to clipping; its variance ratio lies between 0.78 and 1.41. These are point estimates: with 16 draws its bootstrap bounds stay within the lines on 1 of them.

Table 5: Revised main table (repaired simulator): gap to digital Adam (its own selected learning rate), mean [95 % t-interval] over 5 seeds paired by seed (same initialisation and data order); MNIST and CIFAR in accuracy points, LM in % of bits per byte; P/F/U = PASS/FAIL/UNRESOLVED against the absolute lines 2 pp, 2 pp, 5 %.

Device	Alg.	MNIST	CIFAR	LM
Li-ECRAM	SGD	6.39 [3.79, 8.98] F	20.25 [14.51, 26.00] F	90.33 [60.59, 120.08] F
Li-ECRAM	TTv2	0.42 [0.21, 0.63] P	9.72 [8.82, 10.62] F	35.95 [34.87, 37.02] F
Li-ECRAM	c-TTv2	0.60 [0.26, 0.93] P	0.86 [0.17, 1.55] P	23.49 [22.61, 24.36] F
Li-ECRAM	AGAD	0.54 [0.26, 0.82] P	0.23 [-0.38, 0.84] P	23.89 [22.95, 24.82] F
MO-ECRAM	SGD	11.48 [4.73, 18.24] F	28.31 [20.59, 36.03] F	117.64 [35.41, 199.88] F
MO-ECRAM	TTv2	1.90 [1.77, 2.04] U	8.11 [7.85, 8.38] F	49.48 [47.97, 50.98] F
MO-ECRAM	c-TTv2	1.01 [0.96, 1.06] P	1.40 [0.27, 2.53] U	39.47 [37.39, 41.55] F
MO-ECRAM	AGAD	0.78 [0.60, 0.96] P	0.28 [-0.30, 0.86] P	35.33 [31.47, 39.19] F
Protonic-inspired	SGD	4.06 [2.18, 5.94] F	13.73 [10.41, 17.05] F	70.37 [65.57, 75.16] F
Protonic-inspired	TTv2	0.54 [0.23, 0.86] P	7.15 [5.27, 9.04] F	31.98 [30.89, 33.06] F
Protonic-inspired	c-TTv2	0.27 [0.16, 0.38] P	0.70 [-1.68, 3.07] U	23.36 [22.15, 24.58] F
Protonic-inspired	AGAD	0.72 [0.58, 0.86] P	-0.27 [-1.30, 0.75] P	25.23 [23.43, 27.03] F
W/WO$_x$ ECRAM	SGD	12.03 [6.56, 17.50] F	34.07 [28.80, 39.34] F	90.44 [82.88, 97.99] F
W/WO$_x$ ECRAM	TTv2	1.44 [1.23, 1.66] P	13.60 [11.76, 15.44] F	53.46 [52.68, 54.24] F
W/WO$_x$ ECRAM	c-TTv2	1.50 [1.39, 1.60] P	11.74 [10.96, 12.52] F	49.10 [47.30, 50.90] F
W/WO$_x$ ECRAM	AGAD	1.04 [0.66, 1.42] P	13.79 [12.70, 14.88] F	52.40 [51.04, 53.75] F
Memistor (optimistic)	SGD	1.58 [0.75, 2.41] U	7.12 [4.95, 9.30] F	57.52 [53.05, 61.99] F
Memistor (optimistic)	TTv2	0.88 [0.57, 1.19] P	2.46 [0.65, 4.28] U	29.93 [28.82, 31.03] F
Memistor (optimistic)	c-TTv2	0.16 [0.01, 0.30] P	1.51 [-0.47, 3.49] U	29.80 [28.39, 31.22] F
Memistor (optimistic)	AGAD	0.60 [0.35, 0.84] P	1.09 [-0.24, 2.43] U	29.99 [28.42, 31.56] F
Ideal 1000-level	SGD	-0.02 [-0.22, 0.18] P	7.39 [3.31, 11.46] F	40.53 [38.35, 42.71] F
Ideal 1000-level	TTv2	0.81 [0.54, 1.07] P	5.79 [0.66, 10.92] U	49.79 [48.21, 51.37] F
Ideal 1000-level	c-TTv2	0.27 [0.10, 0.44] P	19.10 [10.33, 27.88] F	43.45 [41.48, 45.41] F
Ideal 1000-level	AGAD	0.75 [0.47, 1.03] P	20.27 [15.32, 25.22] F	40.36 [38.70, 42.01] F

- It fails on the two probes of the most clipped layer: CIFAR MO-ECRAM AGAD layer 4 at update 600 (95.7 % lost, gain +1.5 %, excess 0.15); CIFAR MO-ECRAM AGAD layer 4 at update 2000 (97.3 % lost, gain -4.2 %, excess 0.30). There almost nothing of the requested update survives the clipping, and the surrogate's Poisson counts, which can exceed the slot budget that caps the stream, no longer match it (review 2's limiting case: Poisson(31) exceeds 31 with probability 0.4524).

- Paired by draw, seq meets its registered lines (gain within 5 %, excess at most 0.05, variance ratio in $[0.8, 1.25]$) on 12 of 12 probes. The paired floor, 0.29–0.98 times the independent one, accounts for the raw relative difference (at most 0.224) up to an excess of at most 0.015; gains agree within 1.4 % and variance ratios lie in 0.993–1.029. The bootstrap bounds stay within the lines on 6 probes (LM layer 0 at update 750; LM layer 0 at update 2250; MNIST layer 0 at update 1000; MNIST layer 2 at update 1000; MNIST layer 0 at update 3500; MNIST layer 2 at update 3500). On the others (CIFAR layer 3 at update 600; CIFAR layer 4 at update 600; CIFAR layer 3 at update 2000; CIFAR layer 4 at update 2000; LM layer 9 at update 750; LM layer 9 at update 2250) 16 draws cannot exclude a mean discrepancy of up to 0.286. Seq therefore meets our registered diagnostic thresholds; that is a diagnostic, not a demonstration that it is equivalent to the literal stream.

- Shared lines make the literal stream's cells dependent; the surrogate's independent draws do not reproduce this. Measured by per-pair correlations, which cannot cancel (Table 8), the

root-mean-square correlation among cells sharing a line exceeds the surrogate's sampling null, in quadrature ($\sqrt{\max(0,\ \mathrm{RMS}^2_{\mathrm{slot}} - \mathrm{RMS}^2_{\mathrm{null}})}$), by up to 0.345 (mean correlation aligned with the expected update's signs up to 0.208), by more than 0.05 on 6 of 12 probes (LM layer 9 at update 750; LM layer 9 at update 2250; MNIST layer 0 at update 1000; MNIST layer 2 at update 1000; MNIST layer 0 at update 3500; MNIST layer 2 at update 3500). The cross-cell dependence is therefore not small everywhere; the per-cell lines above do not test it, and whether it matters for training is what A2 measures.

The probed states are fresh training realisations on the desk's GPU (Appendix F); the comparison within each probe is unaffected, since all three updates start from the same saved state.

Training level (A2). The same selections, learning rates and seeds as the main table were trained with the per-sample stream (MNIST: every analog layer, three devices × SGD, c-TTv2, AGAD; CIFAR: the two fully connected layers, whose rows are one per image, for Li-ECRAM and protonic-inspired c-TTv2 and MO-ECRAM AGAD; the convolutional layers keep the surrogate), seeds 0–2, and compared seed by seed with the main table's surrogate runs; both arms are labelled at the same three seeds (Table 10). The pre-registered rule calls the change material if any cell's mean gap moves by more than half the line (1 pp) or two cells whose gaps differ by more than 1 pp swap order.

- By that rule the result is **material**, through one cell: MNIST MO-ECRAM SGD, whose gap moves from 14.10 to 12.68 pp (difference -1.42 pp, 95 % interval [-4.84, 2.00]). At three seeds both of its arms are UNRESOLVED (surrogate [0.04, 28.15], stream [1.27, 24.09] pp). It was rerun as matched triples on one GPU (experiment A2-M, Table 9; `PLAN-REVISION-4.md`): on the desk's Quadro P2200, each of seeds 0–4 trained the FP-SGD control, the surrogate and the stream from one initialisation and one example order (equal hashes, checked; the analysis refuses a seed without them). At these 5 matched seeds the stream's gap is 8.73 pp [6.82, 10.65], FAIL, against 8.71 pp [7.02, 10.40], FAIL, with the surrogate; the paired change is 0.02 pp [-0.47, 0.52] (not material; its interval lies within ±1 pp). The FAIL label holds under either update. This cell's gap depends strongly on the random realisation (Appendix F), and the update's effect on it is measured only by the paired comparisons within each matched set.

- One label changes at the same three seeds: CIFAR Li-ECRAM c-TTv2 (PASS with the surrogate, UNRESOLVED with the stream). Its mean gap moves by only 0.52 pp, but the stream's interval ([-6.44, 5.20] pp) is wider and includes the line. This is not a failure, but the label is not preserved, and the difference interval ([-3.26, 4.29] pp) does not bound the change within ±1 pp.

- 2 pairs of cells reverse their mean order (MNIST MO-ECRAM AGAD and MNIST Protonic-inspired AGAD (separation -0.02 to 0.12 pp); CIFAR Li-ECRAM c-TTv2 and CIFAR Protonic-inspired c-TTv2 (separation -0.29 to 0.84 pp)); none was separated by more than 1 pp under the surrogate, so there is no material rank reversal under the registered rule, but the ranking as a whole is not preserved. No Tiki-Taka-family cell's mean moves by more than 0.61 pp (MNIST at most 1.42 pp including the SGD cell; CIFAR at most 0.61 pp, with intervals of several pp at three seeds).

Within these bounds (MNIST on every layer, CIFAR on its fully connected layers, the LM at update level only, three seeds), the registered result is one material mean change, in a failing SGD cell whose FAIL label holds at five matched seeds on one GPU under either update; no material

Table 6: Update fidelity of the aggregate surrogate against the literal pulse stream (experiment A1-P; desk rerun of A1 with the draws kept, 16 draws per probe; Li = Li-ECRAM, MO = MO-ECRAM, Prot. = protonic-inspired). Lost = requested pulse mass removed by the probability clipping (%). The surrogate's draws are independent of the stream's, so the independent sampling floor applies: gain difference (%, bootstrap 95 % interval over draws), excess relative mean difference beyond the floor (bootstrap 95th percentile), median variance ratio, and whether the point estimates meet the registered lines (10 %, 0.10, [0.5, 2]). "(95)": the bootstrap 95th percentile of the excess and the bootstrap 95 % gain interval are also within their lines; it does not cover the variance ratio (a point estimate) and is not a joint 95 % guarantee for every line.

Case	L.	Upd.	Lost	Δg %	excess	var.	meets
CIFAR MO AGAD	3	600	36.2	1.4 [1.2, 1.6]	0.06 (0.17)	1.14	yes
CIFAR MO AGAD	4	600	95.7	1.5 [0.5, 2.6]	0.15 (0.16)	1.26	no
CIFAR MO AGAD	3	2000	26.2	0.5 [-0.3, 1.3]	0.03 (0.31)	1.07	yes
CIFAR MO AGAD	4	2000	97.3	-4.2 [-7.8, -0.2]	0.30 (0.32)	1.27	no
LM Prot. c-TTv2	0	750	6.1	1.3 [0.8, 1.8]	0.03 (0.14)	1.01	yes
LM Prot. c-TTv2	9	750	0.0	-0.5 [-2.4, 1.8]	0.03 (0.32)	0.86	yes
LM Prot. c-TTv2	0	2250	7.0	-0.6 [-1.0, -0.1]	0.04 (0.13)	1.05	yes
LM Prot. c-TTv2	9	2250	0.0	0.3 [-1.8, 2.3]	0.04 (0.32)	0.78	yes
MNIST Li c-TTv2	0	1000	16.0	-0.4 [-0.8, 0.1]	0.00 (0.14)	1.06	yes
MNIST Li c-TTv2	2	1000	4.6	-1.1 [-2.6, 0.6]	0.02 (0.10)	1.19	yes
MNIST Li c-TTv2	0	3500	37.5	0.6 [0.0, 1.1]	0.00 (0.15)	1.11	yes
MNIST Li c-TTv2	2	3500	5.7	0.7 [-0.5, 2.1]	0.02 (0.09)	1.41	yes (95)

rank reversal; no Tiki-Taka-family mean shift above 0.61 pp; and one label that loses PASS. This is not a demonstration that the surrogate and the stream train equivalently, and the per-sample stream is itself an approximation of the literal one. The surrogate is not validated where clipping removes nearly all of the update (the CIFAR output layer above), for the convolutional layers, or for the LM's training trajectory, and the main results remain comparisons within this aggregate-update simulator.

4.4 Width

An earlier lane of this project saw the device gap grow with model width. Here every arm (protonic-inspired and Li-ECRAM, each with TTv2 and c-TTv2) gets its own learning rate at every width by the edge rule on validation, and then five test seeds per width; FP-SGD uses its own selected rate (Table 11; the rates are in Table 14). As in the rest of the paper, each layer's weight scale α is set from its initialisation bound, so it falls as $1/\sqrt{\text{fan-in}}$ when the width raises a layer's fan-in (layers with a fixed fan-in keep theirs); this is one mapping policy, not every fixed physical mapping.

The test. The endpoint contrast D (the mean gap at the widest minus the narrowest width) is tested for equivalence within the gap lines (TOST at 5 %). The pre-registered contrast is Welch's, which treats the two widths' seed samples as independent; its interval takes the t quantile at the Welch–Satterthwaite degrees of freedom (deviation R9; Appendix F). The samples are not independent: a seed keeps its example order across widths, and within a width every arm of a seed also shares the logical initialisation (between widths the layers differ in shape). Beside the registered result we therefore report the contrast paired by seed, with the same decision order, as a sensitivity analysis added after the second review (deviation R4). Both test the endpoints only; neither covers the middle width.

Table 7: Update fidelity of the per-sample stream (seq) against the literal pulse stream, paired by draw (shared line firings; experiment A1-P, the probes of Table 6): gain difference (%, bootstrap 95 % interval), raw relative mean difference, independent / paired sampling floor, excess beyond the PAIRED floor (bootstrap 95th percentile), median variance ratio, and whether the point estimates meet the registered lines (5 %, 0.05, [0.8, 1.25]). "(95)": the bootstrap 95th percentile of the excess and the bootstrap 95 % gain interval are also within their lines; it does not cover the variance ratio (a point estimate) and is not a joint 95 % guarantee for every line.

Case	L.	Upd.	Δg %	raw	floor ind./pair.	excess	var.	meets
CIFAR MO AGAD	3	600	0.1 [-0.1, 0.2]	0.114	0.121 / 0.114	0.011 (0.144)	1.00	yes
CIFAR MO AGAD	4	600	-0.9 [-1.5, -0.3]	0.050	0.050 / 0.050	0.005 (0.064)	1.03	yes
CIFAR MO AGAD	3	2000	-0.2 [-0.6, 0.1]	0.224	0.241 / 0.224	0.015 (0.286)	1.00	yes
CIFAR MO AGAD	4	2000	-1.4 [-5.5, 3.5]	0.072	0.074 / 0.073	0.000 (0.094)	1.00	yes
LM Prot. c-TTv2	0	750	-0.0 [-0.2, 0.2]	0.032	0.106 / 0.032	0.000 (0.041)	1.00	yes (95)
LM Prot. c-TTv2	9	750	0.0 [-0.3, 0.3]	0.075	0.258 / 0.075	0.000 (0.095)	1.00	yes
LM Prot. c-TTv2	0	2250	0.1 [-0.0, 0.3]	0.029	0.093 / 0.028	0.004 (0.037)	1.00	yes (95)
LM Prot. c-TTv2	9	2250	0.2 [-0.2, 0.5]	0.073	0.249 / 0.073	0.007 (0.093)	1.00	yes
MNIST Li c-TTv2	0	1000	-0.0 [-0.0, 0.0]	0.036	0.109 / 0.036	0.001 (0.045)	1.00	yes (95)
MNIST Li c-TTv2	2	1000	0.1 [-0.1, 0.2]	0.027	0.074 / 0.027	0.000 (0.035)	1.02	yes (95)
MNIST Li c-TTv2	0	3500	-0.0 [-0.1, 0.0]	0.036	0.113 / 0.036	0.003 (0.046)	1.00	yes (95)
MNIST Li c-TTv2	2	3500	0.1 [-0.2, 0.3]	0.024	0.063 / 0.025	0.000 (0.031)	0.99	yes (95)

CIFAR ($c = 16$, 32, 64). 1 arm is equivalent within ±2 pp (Li-ECRAM c-TTv2, $D = -0.34$ pp, 90 % interval [-1.34, 0.65]); for 1 arm there is no evidence of growth beyond the margin (protonic-inspired TTv2); 1 arm grows (protonic-inspired c-TTv2, $D = 1.76$ pp, 90 % interval [0.05, 3.47]), and 1 is inconclusive (Li-ECRAM TTv2, $D = 1.91$ pp). Five seeds are not enough to settle a 2 pp margin on this subset. Paired by seed, the Li-ECRAM c-TTv2 equivalence survives (90 % interval [-1.79, 1.11]), and protonic-inspired c-TTv2 grows again ([0.15, 3.38]). Its growth is detected by both contrasts, but its size against the margin is unresolved: both intervals include 2 pp, so growth beyond the margin is not excluded. Both of its endpoint cells still pass against their own FP-SGD controls; a gap that moves from more negative to less negative can grow while both cells stay acceptable. Li-ECRAM TTv2 stays inconclusive, so growth beyond the margin is not excluded for it either.

The LM ($d = 64$, 128, 256). All 4 arms have a gap that shrinks with width (for example Li-ECRAM c-TTv2 from 1.21 % to -12.90 %). This is not a favourable scaling law: FP-SGD itself gets worse at the largest width under the fixed budget (2.962 bits per byte at $d = 128$, 3.117 at $d = 256$), so a gap that turns negative partly measures a weaker baseline. Converged or quality-matched baselines would be needed to speak of scaling.

What can be said. Across a factor of four in width, after per-arm learning-rate selection, no arm's gap grows detectably beyond the margin. Endpoint equivalence is established for one CIFAR arm only (Li-ECRAM c-TTv2, by both contrasts); both contrasts detect growth for one other CIFAR arm (protonic-inspired c-TTv2), without resolving its size against the margin; for two arms (protonic-inspired c-TTv2 and Li-ECRAM TTv2) growth beyond the margin is not excluded. On the LM the apparent shrinkage reflects the FP-SGD baseline at a fixed budget. Width dependence is unresolved by this study.

Table 8: Cross-cell dependence of the literal stream's draws (experiment A1-P), among cells sharing a row / a column, from per-pair Pearson correlations over all pairs (they cannot cancel): root-mean-square correlation of the literal stream and of the surrogate (whose cells are independent, so its value is the sampling null at the same number of draws), the excess RMS beyond that null, taken in quadrature ($\sqrt{\max(0,\ \mathrm{RMS}^2_{\mathrm{slot}} - \mathrm{RMS}^2_{\mathrm{null}})}$), the mean correlation aligned with the signs of the expected update, the largest per-line RMS, and the version-3 statistic (a pooled, signed normalised covariance, which can cancel).

Case	L.	Upd.	RMS slot	RMS sur. (null)	excess RMS	aligned mean	max line RMS	v3 pooled
CIFAR MO AGAD	3	600	0.258 / 0.261	0.258 / 0.258	0.009 / 0.041	0.005 / 0.010	0.262 / 1.000	0.000 / -0.000
CIFAR MO AGAD	4	600	0.256 / 0.261	0.259 / 0.260	0.000 / 0.026	0.007 / -0.004	0.259 / 0.645	0.000 / -0.008
CIFAR MO AGAD	3	2000	0.260 / 0.260	0.258 / 0.258	0.032 / 0.026	0.007 / 0.004	0.283 / 0.399	-0.000 / -0.000
CIFAR MO AGAD	4	2000	0.259 / 0.260	0.259 / 0.255	0.015 / 0.048	0.006 / -0.003	0.265 / 0.315	0.000 / 0.001
LM Prot. c-TTv2	0	750	0.259 / 0.258	0.258 / 0.258	0.015 / 0.008	-0.000 / 0.000	0.266 / 0.264	0.000 / 0.000
LM Prot. c-TTv2	9	750	0.299 / 0.258	0.258 / 0.258	0.152 / 0.007	0.026 / -0.000	1.000 / 0.264	-0.000 / -0.001
LM Prot. c-TTv2	0	2250	0.258 / 0.258	0.258 / 0.258	0.000 / 0.000	-0.000 / 0.000	0.265 / 0.262	0.000 / -0.000
LM Prot. c-TTv2	9	2250	0.315 / 0.258	0.258 / 0.258	0.180 / 0.003	0.025 / -0.000	1.000 / 0.266	0.001 / -0.001
MNIST Li c-TTv2	0	1000	0.335 / 0.262	0.258 / 0.258	0.213 / 0.044	0.054 / 0.010	0.454 / 0.589	-0.000 / -0.000
MNIST Li c-TTv2	2	1000	0.399 / 0.262	0.258 / 0.260	0.305 / 0.031	0.103 / 0.014	0.494 / 0.348	-0.000 / -0.020
MNIST Li c-TTv2	0	3500	0.385 / 0.259	0.258 / 0.258	0.286 / 0.025	0.100 / 0.004	0.478 / 0.322	-0.001 / -0.000
MNIST Li c-TTv2	2	3500	0.431 / 0.264	0.258 / 0.258	0.345 / 0.057	0.208 / 0.003	0.522 / 0.321	0.006 / -0.003

4.5 Exploratory device sensitivity against absolute lines

Around the protonic-inspired base we move one device property at a time (levels, step asymmetry, saturation γ, write noise, reference error, retention) at the base sizes (CIFAR $c = 32$, LM $d = 128$), three seeds paired with FP-SGD of the same seed, at FP-SGD's learning rate. Every grid point is labelled against the *absolute* line (2 pp, 5 %), never against the base device's own gap (Table 12). The asymmetry sweep is run twice: as before, where the symmetry point $u^* = a/(2\gamma)$ moves toward the upper bound and the positive weight range shrinks ($a = 0.4$ at $\gamma = 0.2$ puts u^* on the bound); and with each device's range re-centred on its symmetry point, so that only the up/down dynamics change.

The pre-registered sheet (TTv2) establishes no threshold. Its base point is itself UNRESOLVED on both tasks (6.35 pp on CIFAR, 5.15 % on the LM), so by the pre-registered rule every axis is "not established". It still shows the confound the review pointed out: at $a = 0.4$ the gap is 42.92 pp on CIFAR when the symmetry point moves onto the bound, but 6.20 pp when it is held at mid-range. The earlier "asymmetry limit" was mostly lost weight range.

The chopped-TTv2 sheet (added after the main table, deviation R3). On CIFAR the base passes (-0.85 pp, PASS), and the sheet establishes: step asymmetry up to 0.2 with the symmetry point moving (at 0.4 the gap jumps to 42.39 pp, when u^* reaches the bound), and reference error up to 0.05 of the half range. With the symmetry point held at mid-range, the base and $a = 0$ are run with each device's range re-centred (range $[-1, 1]$; pre-registered after the second review, deviation R5, Appendix F). The recentred base passes (-0.41 pp) and so does $a = 0$ (-1.43 pp), so the held-symmetry threshold 0.05 is now established by its own runs; it differs from the unrecentred base by 0.44 pp (95 % interval [-2.66, 3.54]). No point fails: up to $a = 0.4$ the mean gap stays at -0.53 pp and the intervals merely include the line. Levels, saturation, write noise and finite retention are not established: neighbouring grid values are UNRESOLVED with three seeds, while the clear failures are no saturation ($\gamma = 0$: 21.14 pp), strong saturation ($\gamma = 1$: 8.01 pp), reference error 0.2

Table 9: Experiment A2-M: MNIST MO-ECRAM SGD as matched triples on one GPU (Quadro P2200): per seed, the FP-SGD control's, the surrogate's and the per-sample stream's test accuracy (%) and the two gaps to the control (pp); every seed's three runs share their initialisation and example order (equal hashes, checked). Right: the same seeds' gaps from the pod's matched triples (RTX A4000), a different random realisation of each seed.

	P2200					pod (A4000)	
Seed	FP-SGD	sur.	seq	gap sur.	gap seq	gap sur.	gap seq
0	97.90	89.83	90.02	8.07	7.88	16.84	15.69
1	97.60	87.86	87.30	9.74	10.30	7.59	7.39
2	97.68	89.79	90.26	7.89	7.42	17.86	14.95
3	97.74	87.20	87.22	10.54	10.52	–	–
4	97.75	90.44	90.20	7.31	7.55	–	–
mean [95 %]				8.71 [7.02, 10.40]	8.73 [6.82, 10.65]		

Table 10: Training fidelity (experiment A2), seeds 0–2 on both arms: mean gap to FP-SGD (pp) with its 95 % interval and label against the 2 pp line (P pass, U unresolved, F fail), with the aggregate surrogate (E3's records) and with the per-sample pulse stream (MNIST: every analog layer; CIFAR: the two fully connected layers); the paired difference with its 95 % interval; material by the registered rule (|mean| > 1 pp). The labels are at $n = 3$; the main table's labels are at $n = 5$.

Task	Cell	surrogate	stream	difference	material
CIFAR	Li-ECRAM c-TTv2	-1.14 [-3.89, 1.62] P	-0.62 [-6.44, 5.20] U	0.52 [-3.26, 4.29]	no
CIFAR	MO-ECRAM AGAD	-1.40 [-3.22, 0.41] P	-1.82 [-3.76, 0.13] P	-0.41 [-1.99, 1.16]	no
CIFAR	Protonic-inspired c-TTv2	-0.85 [-3.33, 1.64] P	-1.46 [-3.94, 1.02] P	-0.61 [-2.86, 1.64]	no
MNIST	Li-ECRAM AGAD	0.65 [-0.26, 1.56] P	0.49 [0.08, 0.90] P	-0.16 [-0.69, 0.36]	no
MNIST	Li-ECRAM c-TTv2	0.68 [-0.77, 2.12] U	0.74 [-1.70, 3.18] U	0.07 [-0.95, 1.09]	no
MNIST	Li-ECRAM SGD	6.99 [0.52, 13.46] U	6.90 [1.03, 12.76] U	-0.09 [-0.73, 0.54]	no
MNIST	MO-ECRAM AGAD	0.77 [0.48, 1.06] P	0.89 [0.70, 1.07] P	0.11 [-0.07, 0.30]	no
MNIST	MO-ECRAM c-TTv2	1.08 [0.44, 1.72] P	1.05 [0.43, 1.66] P	-0.04 [-0.23, 0.16]	no
MNIST	MO-ECRAM SGD	14.10 [0.04, 28.15] U	12.68 [1.27, 24.09] U	-1.42 [-4.84, 2.00]	yes
MNIST	Protonic-inspired AGAD	0.79 [0.15, 1.43] P	0.77 [0.17, 1.37] P	-0.02 [-0.39, 0.35]	no
MNIST	Protonic-inspired c-TTv2	0.29 [-0.30, 0.88] P	0.21 [-0.71, 1.13] P	-0.08 [-0.44, 0.27]	no
MNIST	Protonic-inspired SGD	4.58 [0.15, 9.00] U	4.69 [0.01, 9.38] U	0.12 [-0.33, 0.56]	no

(9.31 pp) and retention of 10^3 updates or less. On the LM the base is UNRESOLVED with three seeds (-1.84 %; it passed with five seeds in the main table), so no LM threshold is established.

How to read this analysis. It is an exploratory device sensitivity analysis, not a specification sheet: one-at-a-time ranges around one base device, for one algorithm setting, at FP-SGD's learning rate, with three seeds, and not fabrication tolerances. A range established for one property does not bound joint deviations: passing with a step asymmetry alone and with a reference error alone does not show passing with both. Other algorithm settings and independent device populations are not covered either. A near-zero saturation fails here because, with these settings, the accumulator's decay comes from the device (Section 4.2); it is not a general requirement for training. The rule is deliberately stringent: an UNRESOLVED point at the best end of an axis blocks a threshold even where intermediate points pass. "Not established" therefore does not mean that no grid value passed (on CIFAR with c-TTv2 the 500, 1000 and 2000 level points pass, -1.57, -0.85 and -2.43 pp, while 5000 levels give 2.49 pp, UNRESOLVED), and it does not show that more levels

Table 11: Revised width study (Li = Li-ECRAM, Prot. = protonic-inspired): mean gap at the narrow / mid / wide width (5 seeds, each arm at its own selected learning rate; CIFAR $c = 16/32/64$ in pp, LM $d = 64/128/256$ in %), and the endpoint contrast D (widest minus narrowest) with its 90 % interval and verdict, by the pre-registered Welch test (the t quantile at the Welch–Satterthwaite degrees of freedom; versions 2–11 rounded them down: deviation R9) and, added after review 2 (deviation R4), paired by seed. Verdicts (margins 2 pp, 5 %): EQ = equivalent at the endpoints, GR = grows, SH = shrinks, NG = no evidence of growth beyond the margin, INC = inconclusive. An endpoint test says nothing about the middle width.

Task	Arm	gap narrow / mid / wide	Welch D [90 %]	paired D [90 %]
CIFAR	Li c-TTv2	-0.36 / 0.66 / -0.71	-0.34 [-1.34, 0.65] EQ	-0.34 [-1.79, 1.11] EQ
CIFAR	Li TTv2	3.29 / 4.12 / 5.20	1.91 [-0.34, 4.16] INC	1.91 [-1.53, 5.35] INC
CIFAR	Prot. c-TTv2	-2.57 / -1.55 / -0.81	1.76 [0.05, 3.47] GR	1.76 [0.15, 3.38] GR
CIFAR	Prot. TTv2	1.38 / 2.71 / 0.43	-0.95 [-2.51, 0.60] NG	-0.95 [-3.36, 1.46] NG
LM	Li c-TTv2	1.21 / -4.69 / -12.90	-14.12 [-15.37, -12.87] SH	-14.12 [-15.30, -12.94] SH
LM	Li TTv2	9.85 / 13.29 / -3.07	-12.92 [-14.55, -11.29] SH	-12.92 [-14.99, -10.85] SH
LM	Prot. c-TTv2	-5.79 / -6.36 / -12.69	-6.90 [-8.08, -5.72] SH	-6.90 [-7.76, -6.04] SH
LM	Prot. TTv2	7.95 / 0.39 / -6.49	-14.44 [-19.12, -9.75] SH	-14.44 [-19.31, -9.57] SH

harm hardware training in general.

Table 12: Revised sensitivity sheet (protonic-inspired base; TTv2 = the pre-registered S-ALGO, c-TTv2 added after the main table was seen, deviation R3; 3 seeds paired with FP-SGD): mean gap per grid value with its label against the ABSOLUTE line (P/F/U), and the weakest value that passes with every value toward the best end also passing (– = not established).

Alg., task	Axis	mean gap per value	threshold
c-TTv2, CIFAR	levels	5000.0: 2.49U; 2000.0: -2.43P; 1000: -0.85P; 500.0: -1.57P; 200.0: 1.38U; 100.0: 1.68U	–
c-TTv2, CIFAR	asym. a (SP moves)	0.0: -1.43P; 0.05: -0.85P; 0.1: -1.64P; 0.2: -1.16P; 0.4: 42.39F	0.2
c-TTv2, CIFAR	asym. a (SP fixed)	0.0: -1.43P; 0.05: -0.41P; 0.1: -0.82U; 0.2: -1.38U; 0.4: -0.53U	0.05
c-TTv2, CIFAR	nonlin. γ	0.0: 21.14F; 0.2: -0.85P; 0.5: 1.88U; 1.0: 8.01F; 2.0: 14.77F	–
c-TTv2, CIFAR	noise c_{2c}	0.0: -0.64U; 0.3: -0.85P; 1.0: -1.34U; 2.0: 0.61U; 4.0: 5.37U	–
c-TTv2, CIFAR	ref. err. σ_r	0.0: -1.20P; 0.02: -0.85P; 0.05: -1.29P; 0.1: 2.89U; 0.2: 9.31F; 0.5: 34.63F	0.05
c-TTv2, CIFAR	retention τ	inf: -0.85P; 100000.0: -1.17U; 10000.0: 3.08U; 1000.0: 33.02F; 100.0: 44.03F	inf
c-TTv2, LM	levels	5000.0: 2.01U; 2000.0: -2.63U; 1000: -1.84U; 500.0: -0.45U; 200.0: 1.58U; 100.0: 6.54U	–
c-TTv2, LM	asym. a (SP moves)	0.0: -1.68U; 0.05: -1.84U; 0.1: -1.54U; 0.2: -1.85U; 0.4: 37.71U	–
c-TTv2, LM	asym. a (SP fixed)	0.0: -1.62U; 0.05: -2.16U; 0.1: -1.69U; 0.2: -1.70U; 0.4: -1.68U	–
c-TTv2, LM	nonlin. γ	0.0: 22.71F; 0.2: -1.84U; 0.5: -0.39U; 1.0: 4.46U; 2.0: 27.27U	–
c-TTv2, LM	noise c_{2c}	0.0: -1.51U; 0.3: -1.84U; 1.0: -0.21U; 2.0: 4.91U; 4.0: 19.04F	–
c-TTv2, LM	ref. err. σ_r	0.0: -2.42U; 0.02: -1.84U; 0.05: 1.45U; 0.1: 8.23U; 0.2: 17.53F; 0.5: 30.83F	–
c-TTv2, LM	retention τ	inf: -1.84U; 100000.0: -1.75U; 10000.0: -3.47U; 1000.0: 28.50F; 100.0: 64.34F	–
TTv2, CIFAR	levels	5000.0: 0.78U; 2000.0: 1.67U; 1000: 6.35U; 500.0: 4.43U; 200.0: 3.13U; 100.0: 3.12U	–
TTv2, CIFAR	asym. a (SP moves)	0.0: 5.92U; 0.05: 6.35U; 0.1: 5.95F; 0.2: 6.39F; 0.4: 42.92F	–
TTv2, CIFAR	asym. a (SP fixed)	0.0: 5.92U; 0.05: 6.02U; 0.1: 6.12U; 0.2: 4.60U; 0.4: 6.20U	–
TTv2, CIFAR	nonlin. γ	0.0: 27.43F; 0.2: 6.35U; 0.5: 9.00F; 1.0: 8.22F; 2.0: 12.20F	–
TTv2, CIFAR	noise c_{2c}	0.0: 5.79U; 0.3: 6.35U; 1.0: 6.19U; 2.0: 7.18U; 4.0: 9.40F	–
TTv2, CIFAR	ref. err. σ_r	0.0: 5.08U; 0.02: 6.35U; 0.05: 5.35U; 0.1: 9.94F; 0.2: 18.74F; 0.5: 30.32F	–
TTv2, CIFAR	retention τ	inf: 6.35U; 100000.0: 4.49U; 10000.0: 5.18U; 1000.0: 40.83F; 100.0: 47.25F	–
TTv2, LM	levels	5000.0: 7.33U; 2000.0: 4.67U; 1000: 5.15U; 500.0: 5.87U; 200.0: 7.38U; 100.0: 9.64U	–
TTv2, LM	asym. a (SP moves)	0.0: 5.26U; 0.05: 5.15U; 0.1: 5.21U; 0.2: 5.00U; 0.4: 10.38U	–
TTv2, LM	asym. a (SP fixed)	0.0: 5.25U; 0.05: 5.15U; 0.1: 5.28U; 0.2: 5.29U; 0.4: 5.75U	–
TTv2, LM	nonlin. γ	0.0: 32.24F; 0.2: 5.15U; 0.5: 11.13U; 1.0: 18.03F; 2.0: 25.84F	–
TTv2, LM	noise c_{2c}	0.0: 5.14U; 0.3: 5.15U; 1.0: 5.56U; 2.0: 7.85U; 4.0: 15.03U	–
TTv2, LM	ref. err. σ_r	0.0: 4.59U; 0.02: 5.15U; 0.05: 7.27U; 0.1: 11.43U; 0.2: 19.66F; 0.5: 31.81F	–
TTv2, LM	retention τ	inf: 5.15U; 100000.0: 5.09U; 10000.0: 3.74U; 1000.0: 30.34F; 100.0: 71.47F	–

5 A conditional energy projection for the weight-matrix operations

This section is an accounting scenario, not an energy measurement of a training system. It asks how much energy the weight-matrix operations of a training step would take if the analog reads ran at the efficiency of a published analog-AI *inference* chip and the updates cost the published or projected pulse energies, and it compares that with an H100.

Assumptions, stated first.

1. **Different read assumptions for accuracy and energy.** The accuracies of Section 4.2 are simulated with ideal fp32 reads; the energy prices finite-precision analog reads. We have not shown that one operating point gives both the reported accuracy and the assumed read efficiency.

2. **The read efficiency is borrowed from a phase-change inference chip** [37]: up to 12.4 TOPS/W sustained, measured on one chip, and 6.94 TOPS/W, the authors' *projected* full-system figure for their five-chip system: the chips were not integrated with digital processing, whose time and energy the authors estimate from their architecture model, and external components such as system buses and voltage regulators are excluded. We report both. Neither is an ECRAM training array, which also needs backward reads, repeated writes, references and Tiki-Taka state.

3. **The boundary is the weight matrices.** Per weight and per training row: the forward and backward reads (two MACs) and the update (device pulses, pulse lines and pulse generation, and, for the TT family, the transfer's column reads, priced as one ADC conversion and one read-add-write of the digital buffer H per read weight). The pulse reset's extra column reads are priced the same way although they do not update H, a conservative approximation: the buffer allowance stands for the arithmetic that turns each re-read into a correction pulse count, and without it the analog energy would be at most 0.15 % lower. Attention products, nonlinearities, normalisation, activation storage, data movement, control, reference maintenance and static power are not modelled on either side, and the digital exclusions do not cancel the analog ones. The rest of the TT family's digital work, called the digital TT state below (the storage of H, AGAD's averages p and p^{ref} and their arithmetic, the chopper signs, the threshold tests and the transfer scheduling), is outside the ratio.

4. **The GPU side** is H100 board power P over its dense BF16 peak F [44] at the utilisation u of a very large training run [45], with 6 FLOP per weight per row (2 each for the forward pass, the input gradient and the weight gradient): $E_{\text{GPU}} = 6P/(Fu)$, with $F = 989.5$ TFLOP/s (half the 1979 TFLOP/s sparsity figure), P from 500 W (our estimate of a sustained draw) to 700 W (the rated maximum) and u from 38 to 43 %. At 700 W and $u = 40.5$ % this is 10.48 pJ per weight per row; over both ranges it is 7.05 pJ (500 W, 43 %) to 11.17 pJ (700 W, 38 %), and with u over the 30–55 % range of our original plan 5.51–14.15 pJ.

Projection. With the repaired pulse counts (resets counted as applied, the pulse reset's extra reads included), the GPU-to-analog ratio per weight per row is 31.5–32.4× at the 12.4 TOPS/W chip figure and 17.9–18.1× at the 6.94 TOPS/W system figure, for every published device and algorithm. The reads are at least 97.0 % of the analog total: the ratio is set almost entirely by the two borrowed efficiencies, and changes in the simulated pulse counts barely move it. It is therefore

Table 13: Conditional energy ratios for cells that pass the stated non-inferiority margin under ideal reads: the GPU-to-analog energy ratio per weight per row (12.4 TOPS/W measured chip figure / 6.94 TOPS/W projected system figure of the same inference chip, which includes estimated digital processing), printed only where the cell PASSES against the baseline; fail = the cell FAILS the margin; n.e. = the criterion is not established (UNRESOLVED). No finite-read operating point is demonstrated.

		MNIST		CIFAR		LM	
Device	Alg.	FP-SGD	Adam	FP-SGD	Adam	FP-SGD	Adam
Li-ECRAM	SGD	fail	fail	fail	fail	fail	fail
Li-ECRAM	TTv2	32 / 18	32 / 18	fail	fail	n.e.	fail
Li-ECRAM	c-TTv2	32 / 18	32 / 18	32 / 18	32 / 18	32 / 18	fail
Li-ECRAM	AGAD	32 / 18	32 / 18	32 / 18	32 / 18	32 / 18	fail
MO-ECRAM	SGD	fail	fail	fail	fail	fail	fail
MO-ECRAM	TTv2	n.e.	n.e.	fail	fail	fail	fail
MO-ECRAM	c-TTv2	32 / 18	32 / 18	n.e.	n.e.	fail	fail
MO-ECRAM	AGAD	32 / 18	32 / 18	32 / 18	32 / 18	fail	fail
Protonic-inspired	SGD	fail	fail	fail	fail	fail	fail
Protonic-inspired	TTv2	32 / 18	32 / 18	fail	fail	n.e.	fail
Protonic-inspired	c-TTv2	32 / 18	32 / 18	32 / 18	n.e.	32 / 18	fail
Protonic-inspired	AGAD	32 / 18	32 / 18	32 / 18	32 / 18	32 / 18	fail
W/WO$_x$ ECRAM	SGD	fail	fail	fail	fail	fail	fail
W/WO$_x$ ECRAM	TTv2	32 / 18	32 / 18	fail	fail	fail	fail
W/WO$_x$ ECRAM	c-TTv2	32 / 18	32 / 18	fail	fail	fail	fail
W/WO$_x$ ECRAM	AGAD	32 / 18	32 / 18	fail	fail	fail	fail

a conditional, read-dominated figure per operation (one forward read, one backward read and one update of each weight per row), not the energy to train a model to a given quality.

Conditional ratios for cells that pass the margin. A low energy per operation is worth nothing without the accuracy. PASS means that the whole confidence interval of the gap lies within the stated non-inferiority margin under ideal reads, not that the losses are equal; we therefore print the ratio only for cells that PASS against the stated baseline (Table 13), mark FAIL cells as missing the margin and UNRESOLVED cells as "criterion not established":

- against FP-SGD: 11, 5 and 4 of the 16 published-device cells per task (MNIST, CIFAR, LM);
- against Adam: 11, 4 and 0 cells. **On the LM no analog cell passes the margin against Adam, so no conditional ratio exists against the stronger digital baseline.**

The comparison is at a fixed training budget; reaching a quality at a different number of steps is not modelled. No joint finite-read operating point of accuracy and energy, and no energy-to-target training trajectory, is demonstrated; finite precision, tiling and utilisation, reference maintenance, the digital TT state, update scheduling and full-system costs stay outside the ratio.

What 100× would need. At the central GPU figure, a 100× ratio needs reads at about 38.6–42.2 TOPS/W including their periphery, several times the chip figure above.

The memistor. Widrow's macroscopic memistor moves tens of microjoules of plating charge per level; its ratio is at most 8.0×10^{-4} (far worse than the GPU), as expected for wet, centimetre-scale cells.

6 Discussion

What the repaired simulations say, within their limits.

- **Against FP-SGD, two device specifications reach the line on all three tasks** with a chopped Tiki-Taka algorithm: the AIHWKIT Li-ECRAM fit and the protonic-inspired scenario, each with c-TTv2 or AGAD (4 pairings).
- **Against Adam, none does.** On the language model the best analog cell is 23.36 % worse in bits per byte; the analog results inherit FP-SGD's optimiser gap (25.35 % on this model). Closing it would need analog-compatible adaptive updates, which this study does not test.
- **Plain SGD at the FP-SGD learning rate** exceeded the margin on every published device and task we ran (12 of 12 cells). This is a statement about that protocol, not about SGD after fair tuning.
- **The algorithm settings use the device.** On an ideal symmetric 1000-level device the selected TT settings lose far more than on the saturating devices, so in this configuration the accumulator's decay comes from the device's saturation.
- **Width** is not resolved: after per-arm tuning no arm's gap grows detectably beyond the margin over a factor of four, endpoint equivalence was established for one CIFAR arm only, the registered Welch test and a paired-by-seed analysis both detect growth for another without resolving its size against the margin (growth beyond it is not excluded there or for Li-ECRAM TTv2), and the LM's shrinking gap reflects a weaker FP-SGD baseline at the largest width.
- **Energy** is a conditional projection set almost entirely by borrowed read efficiencies, and it is reported only for cells that pass the stated margin under ideal reads; against Adam on the LM no cell does.

Widrow's memistor. Taken at its inventor's description (thousands of near-linear levels), an optimistic historical-device scenario attained useful accuracy in this simulator; the pessimistic twin does not. Because its noise, asymmetry, drift and variability are largely unknown, this does not settle the accuracy of the real 1960 device. Its energy per level, tens of microjoules, is many orders of magnitude worse than today's devices.

6.1 Limitations

- **Simulation only.** Every device is a parametric model; the protonic entry is a scenario with estimated noise, asymmetry and saturation, and its finite retention is replaced by no drift. A 2026 measurement on BEOL-compatible protonic ECRAM [8] reports 3 % cycle-to-cycle and 21 % device-to-device variability; we did not refit the catalog to it, and we could not check those values against the full text. Estimated parameters are single points, not propagated ranges.
- **Ideal reads.** Read noise, IR drop, ADC/DAC quantisation and output bounds are not modelled, while the energy projection prices finite-precision reads.
- **The update surrogate.** Pulses of one batch are aggregated and solved as uniformly interleaved; discrete pulses do not commute, per-cell Poisson draws ignore row/column correlations, and the pulse budget clips the TT family's A updates by up to 57.5 % in some

layers. Against explicit pulse streams (Section 4.3) the surrogate meets the registered lines on 10 of 12 update probes (point estimates), fails on the most clipped layer and does not reproduce the dependence of cells that share a line; in short training comparisons on MNIST and CIFAR's fully connected layers one label loses PASS, two closely spaced cells swap order and one failing SGD cell moves by more than 1 pp. The convolutional layers and the LM's training are not compared, and the comparison stream is itself a per-sample approximation of the literal one.

- **The reset.** The pulse reset uses a read and at most 62 pulses per weight; it leaves a residual (mean over runs 67.00 nominal steps, pooled over all reset events 26.97) and its latency and endurance cost are not modelled.
- **Algorithms.** TTv1, TTv2, c-TTv2 and AGAD were implemented from the papers' text; our TTv2 is a local variant (multi-pulse transfer, buffer remainder). The settings were selected once, on MNIST, with a mid device.
- **Controls.** The ideal-device reference keeps stochastic pulses, bounds and the selected update management, so it does not separate the optimiser from the device. A full ladder would run exact digital SGD, a digital TT with an explicit accumulator decay, finite pulses and levels on ideal devices, the device imperfections and shared-line pulse streams; we ran the first, the ideal 1000-level device, the catalog and, on MNIST and CIFAR's dense layers, the per-sample stream.
- **Small models and short budgets.** 10^5 to 10^6 weights, a 10,000-image CIFAR subset, a two-layer byte LM, a factor of four in width, fixed training budgets.
- **Statistics.** Five seeds (three in the sensitivity analysis and the diagnostic), no multiplicity correction, and best cells chosen among many; several cells remain UNRESOLVED.
- **Plans.** The plans were committed to our repository before the runs; they are not registered in an external registry.

7 A proposed roadmap for groups with device and fabrication resources

The simulations suggest what to measure first. None of the criteria below is an established device qualification: each is conditional on this simulator (ideal reads, the pulse surrogate, small models) and on the chopped Tiki-Taka setting that passed in Section 4.2.

Computational follow-ups (no hardware). In order of value: (1) train the passing configurations over full CIFAR and LM trajectories with correlated, shared-line pulse streams on every analog layer (so far compared in training only on MNIST and CIFAR's dense layers, Section 4.3); (2) compare pulse lengths and update-management policies where the update clips (mean lost fraction up to 57.5 %, worst layer record 96.4 %); (3) run the control ladder of Section 6.1; (4) tune each analog arm, plain SGD included, with a search budget comparable to the digital baselines' (the main table reuses the FP-SGD learning rate).

Stage 1: single-device characterisation (tens of devices, no array).

- **Measurement.** At least 1000 up and 1000 down pulses at the intended width and voltage, repeated at least three times on at least 32 devices, with the read current measured (not inferred) and incomplete pulses rejected. Fit: step against state (saturation $\gamma_{\pm}$), up/down asymmetry, cycle-to-cycle noise (separated from read noise), device-to-device spreads, the symmetry point and the signed weight range around it, and drift.
- **What the simulations support** (c-TTv2 sheet, CIFAR $c = 32$, absolute 2 pp line, three seeds, Table 12): a step asymmetry up to 0.2 and a reference error up to 0.05 of the half range, one axis at a time around the protonic-inspired base. Thresholds for levels, saturation, write noise and retention are not established by this grid; what it shows is where training clearly fails (no saturation or $\gamma \geq 1$, reference error 0.2, retention of 10^3 updates or less). A device group should therefore measure all of these and simulate its own fitted device, rather than read pass lines off this sheet. Combined deviations are not covered. Published variability of BEOL-compatible protonic ECRAM [8] (3 % cycle-to-cycle, 21 % device-to-device) is a starting point for such a fit, once its definitions are reconciled with this model's c_{2c} and σ_{d2d}.
- **Endurance and pulse budget.** Report per-array and per-layer pulse distributions and the hottest cells, not only averages; the TT family's A arrays and the resets dominate the write load, and the pulse budget per update clips the largest layers (Section 4.2).

Stage 2: the symmetry point and the reset as physical processes.

- **Measurement.** The pulse count, latency and energy to bring a device to its symmetry point or to its read zero; the residual error, its spread and its drift over hours and 10^6 cycles; the signed range left on each side.
- **Why.** With a realistic pulse budget the reset is incomplete (on the protonic-inspired device, 62 alternating pulses leave 97.55 % of a displacement), so the reset is a device specification, not a free operation.

Stage 3: small arrays with hardware in the loop.

- **Hardware.** Two 64×64 arrays per layer (A and C); forward and backward passes on the host from the read weights; updates on chip as stochastic pulse coincidences; resets or chopping as the algorithm requires.
- **Prediction first.** Before training, fit the Stage 1 model on several independent chips, run this simulator with the fit, and state a *prediction interval* that includes device-population variation, fit uncertainty, training randomness and measurement noise (not a confidence interval of the simulated mean, which shrinks with more seeds). The hardware should land inside it; independently, within 2 pp of FP-SGD shows the device works.

Stage 4: full arrays and in-memory reads. Move the forward and backward reads on chip and measure the system energy per weight per row, with the read precision that gives the accuracy, against the conditional projection of Section 5; convert retention requirements into time with the full step duration.

The 1960 memistor. Appendix A gives a proposed, unbuilt bench replication protocol with hobby parts and safety notes. Before its fits are used as metrology it must measure the actual read current, keep per-pulse charge, separate read from write noise and reject incomplete pulses.

8 Conclusion

In this simulator, with ideal reads and every update made of device pulses, two electrochemical specifications (the AIHWKIT Li-ECRAM fit and a protonic-inspired scenario) train three small networks to within the pre-set margins of SGD with exact updates when a chopped Tiki-Taka algorithm moves the gradient; plain SGD at the same learning rate does not on any published device. Against a tuned digital Adam none of them does, and on the language model the best is 23.36 % worse in bits per byte. After per-arm tuning no arm's gap grows detectably beyond the margin over a factor of four in width; endpoint equivalence was established for one CIFAR pairing only, and both the registered Welch test and a paired-by-seed analysis detect growth for another, of a size not resolved against the margin. The energy advantage is a conditional projection (17.9–32.4× per weight-row at borrowed inference-chip read efficiencies), reported only for cells that pass the margin under ideal reads. After an independent review we repaired the simulator and report how much each repair moved the original results. The remaining questions are computational and methodological (the fidelity of the update surrogate beyond the cases tested here, dependence-aware statistics, the construction of controls) as well as experimental (the reset, the reference and the read path in real arrays).

9 Code and data availability

One public archive holds everything needed to reproduce this paper, with versions, source hashes, dataset digests and split offsets, run manifests and licences: Zenodo, doi:10.5281/zenodo.23748861 (this DOI cites all versions; version 1.5 of the archive holds version 15 of this paper, version 1.4 holds version 14, version 1.3 holds version 13, version 1.2 holds version 12, version 1.1 holds version 11, and version 1.0 holds version 9). The same files are on GitHub under the author's account, `https://github.com/Questeria/memistor-ecram-training` (release v1.5). Code is under the MIT License; records, pulse draws, text and figures are under CC BY 4.0. The archive keeps the study folder's layout, and the paths below are relative to its `paper/` directory. It holds:

- the plans (`PLAN-PAPER.md`, `PLAN-REVISION.md`, `PLAN-REVISION-2.md`, `PLAN-REVISION-3.md`, `PLAN-REVISION-4.md`), their deviations (`DEVIATIONS-PAPER.md`), the twelve independent reviews (`reviews/`) and the responses to them (`RESPONSE-TO-REVIEW.md`, `RESPONSE-TO-REVIEW-2.md`, `RESPONSE-TO-REVIEW-3.md`, `RESPONSE-TO-REVIEW-4.md`, `RESPONSE-TO-REVIEW-5.md`, `RESPONSE-TO-REVIEW-6.md`, `RESPONSE-TO-REVIEW-7.md`, `RESPONSE-TO-REVIEW-8.md`, `RESPONSE-TO-REVIEW-9.md`, `RESPONSE-TO-REVIEW-10.md`, `RESPONSE-TO-REVIEW-11.md`, `RESPONSE-TO-REVIEW-12.md`);
- the original simulator and the repaired one (`sim/*_rev.py`), the explicit pulse streams and probes (`sim/*_rev2.py`, `sim/*_rev3.py`, `sim/sweep_rev4.py`), with their checks;
- the device catalog with per-field provenance;
- every run record (one JSON file per training run, with its configuration, metric, pulse counts and hardware);

- the scripts that derive every number, table and figure of this paper from those records (`analyze.py`, `make_figures.py`, `energy/energy_system.py`; for the revision `revision/analyze_rev.py`, `revision/analyze_d1.py`, `revision/compare_curves.py`, `energy/energy_rev.py`, `revision2/analyze_rev2.py`, `revision3/analyze_rev3.py`, `revision4/analyze_rev4.py`); Figure 1 is drawn by `revision14/make_fig_arch.py` and checked against the selected settings, the records and the simulator by `revision14/check_fig_arch.py`;
- the saved pulse draws of the update probes: 12 NumPy archives (81,905,772 bytes, in `draws/rev3/`; each file's SHA-256 is in its probe record and in `PUBLISHING.md`), from which, with the probe records' metadata (layer, clipping figures, bootstrap counts), `revision4/rebuild_draw_tables.py` recomputes every statistic of Tables 6–8 and checks the rebuilt tables against the paper's byte for byte;
- the citation check (every reference verified against Crossref, the arXiv API or the live URL; `records/CITE-CHECK.json`);
- the bench protocol.

In the archive's copies of the reviews and of one response, local file paths are shortened for privacy; the archive's `REDACTIONS.md` lists every change. Data: MNIST, CIFAR-10 and enwik8 are public and pinned by digest; they are not redistributed, and the archive's `fetch_data.py` downloads them and checks the digests. All text data used for training is human-written (Wikipedia, 2006).

Author contributions and use of AI

Anthony DeMarco conceived and directed the study. Claude (Anthropic), an AI system, wrote the simulator and its checks, designed and pre-registered the experiments under Anthony DeMarco's direction, ran them on rented cloud GPUs, analysed the records, verified the references, and drafted the text. ChatGPT (OpenAI), also an AI system and not an author, reviewed versions 1 to 7 of this paper in turn: it checked the record hashes, independently recomputed the revised statistics, and reported errors, each of which was verified against the files and corrected where confirmed (Appendix F).

Under the policies of arXiv [46] and of most journals, an AI system cannot be an author or take responsibility for a paper. For any formal submission the arXiv-compliant variant of this paper lists Anthony DeMarco as the sole author and states Claude's role in this section. All numbers were produced by scripts from the saved records.

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A Bench replication of the 1960 memistor

This appendix is a protocol for others to run. We built nothing and bought nothing. It is meant to measure, with hobby parts, the properties that Widrow's reports leave open (Section 2): levels, up/down symmetry, linearity, write noise, drift and energy per step. The measured cell can then be placed on the same map as the modern devices. The full protocol, an Arduino sketch, a parts list with estimated prices, and an analysis script that turns the log into a device specification for our simulator are in the repository (`BENCH.md`, `bench/`).

Pass criteria. The cell must show at least 32 near-linear, symmetric levels on one lead: over at least 32 consecutive equal-charge pulses in each direction,

1. the mean up step and the mean down step agree within 10%;
2. the step varies by less than $\pm 30\%$ across the range ($\gamma \leq 0.3$);
3. the pulse-to-pulse scatter is below half a step ($c_{2c} \leq 0.5$).

A four-input ADALINE trained by LMS on four such leads should then cut its mean squared error at least tenfold.

Build.

- Electrolyte: 0.5 M copper(II) sulphate in distilled water. Sulphuric acid and brighteners are deliberately left out for safety, so expect some hysteresis.
- Electrodes: 2 mm HB graphite leads with wrapped-wire contacts, the lower contact insulated with nail polish, a wetted length of about 20 mm, and a bare copper counter electrode.
- Electronics: an Arduino Nano, an INA219 current/voltage sensor, and a CD4052 multiplexer for four-wire reads.

Protocol.

1. Read-noise baseline.
2. Plating range in 1 mC pulses.
3. 64 pulses down and 64 up from mid-range, repeated, to measure the step against conductance, the asymmetry and the scatter.
4. A two-hour (ideally overnight) drift run, covered and then stirred, to separate oxygen-driven corrosion.
5. The integral of cell voltage $\times$ current per pulse, which gives the energy per step.
6. 20 full cycles to measure endurance.
7. The LMS ADALINE.

Safety.

- Copper sulphate is harmful if swallowed, irritates eyes and skin, and is very toxic to aquatic life. Wear splash goggles and nitrile gloves, and work on a tray away from food, children and pets.
- Use low voltage only (USB). Keep electronics out of splash range, keep aluminium away from the solution, and reduce the current if bubbles form.
- Never pour waste down the drain. Take it to household hazardous-waste collection.
- For eye contact, rinse for 15 minutes and seek medical advice.

What the bench cannot show. A pencil-lead memistor is macroscopic. Its energy per step is set by the copper moved, about 15 μJ per level by Widrow's figures and 986 μJ for our pencil-lead estimate. That is many orders of magnitude above an H100's energy per weight update (Section 5). The bench can confirm or refute the linearity and symmetry the memistor promised; it cannot make training cheaper. The cheap direction, if any, is the nanoscale descendants.

B Hyperparameter selection

Learning rates and algorithm settings were selected on validation splits only, with seed 0.

- **MNIST learning rate.** The FP grid was $\{3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}, 10^{-2}\}$, extended by the edge rule. The choice is 0.003.
- **TT-family settings.** They were chosen on MNIST validation with a mid-quality device (128 levels, asymmetry 0.1, γ 0.3, c_{2c} 0.5, σ_r 0.02):
  - TTv2: $\lambda \in \{0.1, 0.3, 1, 3\}$ and a period of 16 or 64 batches, with or without a matched A learning rate; in the revision also with the pulse reset or none;
  - c-TTv2: $\lambda \in \{0.1, 0.3\}$, $\rho \in \{0.05, 0.2\}$, with the pulse reset or none;
  - AGAD: $\lambda \in \{0.1, 0.3\}$, $\rho \in \{0.05, 0.2\}$, $\beta \in \{0.3, 1\}$.
- **Derived choices.** The better chopped algorithm (c-TTv2) is used in the scale study, and the best of the three (TTv2) in the sensitivity analysis.
- **CIFAR and LM learning rates** were chosen per width (Section 4.4); Adam's on a grid of $\{3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}\}$ per task with the edge rule.

Table 14 gives the revised selections that every revised run uses (the original selection table is Table 18, Appendix E).

Table 14: Revised settings (repaired simulator), chosen on MNIST validation with the mid device, seed 0; every E3, E4 and E5 record carries exactly these settings. All TT-family runs use BL 31 pulse slots per sample and at most 31 transfer pulses per weight. Better chopped algorithm: c-TTv2; S-ALGO: TTv2.

Alg.	Setting	A reset	Val. %
TTv2	λ 0.1, A lr $\times$10, period 64	pulse (at most 62 pulses, 2 read-correct iterations)	97.6
c-TTv2	λ 0.1, A lr $\times$10, period 16, ρ 0.05	pulse (at most 62 pulses, 2 read-correct iterations)	97.6
AGAD	λ 0.1, A lr $\times$10, period 16, ρ 0.2, β 0.3	none	97.3

Learning rates. FP-SGD: MNIST 0.003; CIFAR $c = 16/32/64$: 0.0003/0.001/0.0003; LM $d = 64/128/256$: 0.0001/0.0001/3e-05. Adam: MNIST 0.001, CIFAR 0.001, LM 0.003. Width arms (E4), narrow/mid/wide: CIFAR Li-ECRAM c-TTv2: 0.0003/0.003/0.003; CIFAR Li-ECRAM TTv2: 0.0003/0.0003/0.0003; CIFAR Protonic-inspired c-TTv2: 0.003/0.003/0.003; CIFAR Protonic-inspired TTv2: 0.0003/0.0003/0.003; LM Li-ECRAM c-TTv2: 0.0001/0.0003/0.0001; LM Li-ECRAM TTv2: 0.0001/0.0003/0.0001; LM Protonic-inspired c-TTv2: 0.0003/0.0003/0.0001; LM Protonic-inspired TTv2: 0.0003/0.0003/0.0001.

C Simulator checks

Each check runs twice. On the real code it must pass. Once for every planted fault it must fail, which shows that the check catches that fault. A crash under a fault also counts as caught. Both suites ran on the desk before the plan was committed, and again at the start of every cloud session; any failure would have stopped the session.

Table 15: Simulator checks and the planted faults each one catches (records/TESTS-PAPER.json, records/TESTS-laneME-checks-on-paper-sim.json).

Check	Result	Planted faults caught
`c_asym_drift`	pass	asymmetry ignored
`c_c2c_noise`	pass	write noise $\sim$ n not sqrt(n)
`c_constant_step_reference`	pass	constant-step reference at a bound (the D1 bug)
`c_drift`	pass	drift relaxes to 0 not bare
`c_fp_equivalence`	pass	apply_exact sign flipped
`c_level_step`	pass	step size halved
`c_pulse_mean`	pass	pulse slots BL dropped; pulse up/down swapped
`c_symmetry_point`	pass	symmetry point sign
`c_ttv2_learns`	pass	TT transfer sign flipped
`c_family_learns`	pass	A-update inputs not chopped
`c_gamma_d2d_sp`	pass	gamma spread shared by the whole tile
`c_identity_sgd_ttv2`	pass	down-branch saturation uses gamma_up
`c_offset_rejection`	pass	AGAD reference never set; chopper demodulation missing; chopper never flips
`c_ttv1_direction`	pass	TTv1 transfer sign flipped

D Earlier development results

This study began as an exploratory screen with the same simulator (SGD and TTv2 only, 1–3 seeds). The screen is summarised here because some of its observations motivated the pre-registered experiments:

- a CIFAR gap "just over 2 pp";
- a gap that "grows with width";
- a "pencil lead built today" that fails every task.

The earlier runs differ from this paper's in three ways:

- MNIST ran on a different GPU;
- the LM's selection split overlapped its training data (corrected here, Section 3.5);
- CIFAR and the LM had one or two seeds.

With five seeds the earlier point estimates are mostly confirmed (Table 16), but the earlier pass/fail readings were not meaningful at that sample size. The CIFAR baseline's seed-to-seed spread is comparable to the gap line itself.

The pessimistic memistor and the pencil-lead estimate were run only in the earlier screen. Both fail every task with SGD and with TTv2. Their specifications are estimates; the bench protocol of Appendix A would replace the pencil-lead estimate by measurements.

Table 16: Earlier development results (lane ME: same simulator, 1–3 seeds, MNIST on a different GPU, the LM selected on an overlapping split) beside the pre-registered 5-seed values of version 1 of this paper (v1; superseded: the original simulator before its repair; the current values are in Tables 4 and 5). Gaps in pp (MNIST, CIFAR) and % (LM). The pessimistic memistor and the pencil lead appear only in the earlier runs.

		MNIST		CIFAR		LM	
Device	Alg.	earlier	v1	earlier	v1	earlier	v1
Li-ECRAM (Tang 2018)	SGD	5.24 (n=3)	4.85 (n=5)	21.78 (n=1)	20.81 (n=5)	39.75 (n=1)	45.58 (n=5)
Li-ECRAM (Tang 2018)	TTv2	1.17 (n=3)	1.32 (n=5)	2.15 (n=2)	2.95 (n=5)	4.08 (n=2)	3.52 (n=5)
Protonic (Onen 2022)	SGD	3.67 (n=3)	3.07 (n=5)	19.54 (n=1)	15.77 (n=5)	37.62 (n=1)	40.82 (n=5)
Protonic (Onen 2022)	TTv2	0.77 (n=3)	1.04 (n=5)	2.13 (n=2)	2.13 (n=5)	4.21 (n=2)	3.64 (n=5)
MO-ECRAM (Kim 2019)	SGD	7.20 (n=3)	6.96 (n=5)	26.19 (n=1)	25.29 (n=5)	76.49 (n=1)	50.84 (n=5)
MO-ECRAM (Kim 2019)	TTv2	1.84 (n=3)	1.80 (n=5)	8.04 (n=1)	5.45 (n=5)	17.19 (n=1)	16.51 (n=5)
W/YSZ/WO_x (Chen 2023)	SGD	8.19 (n=3)	8.25 (n=5)	37.88 (n=1)	36.82 (n=5)	54.79 (n=1)	50.67 (n=5)
W/YSZ/WO_x (Chen 2023)	TTv2	1.49 (n=3)	1.48 (n=5)	10.45 (n=1)	10.59 (n=5)	20.77 (n=1)	21.17 (n=5)
Memistor (Widrow 1960, as claimed)	SGD	1.71 (n=3)	1.29 (n=5)	17.15 (n=1)	9.86 (n=5)	28.01 (n=1)	30.94 (n=5)
Memistor (Widrow 1960, as claimed)	TTv2	0.63 (n=3)	0.73 (n=5)	3.24 (n=1)	2.86 (n=5)	10.75 (n=1)	8.99 (n=5)
Memistor (pessimistic)	SGD	32.58 (n=3)	–	41.52 (n=1)	–	47.26 (n=1)	–
Memistor (pessimistic)	TTv2	15.33 (n=3)	–	39.16 (n=1)	–	27.44 (n=1)	–
Pencil lead built today	SGD	23.05 (n=3)	–	43.92 (n=1)	–	42.36 (n=1)	–
Pencil lead built today	TTv2	11.72 (n=3)	–	37.39 (n=1)	–	21.98 (n=1)	–

E The original results (superseded)

The first version of this paper reported the experiments below, run on the original simulator. An independent review found that this simulator (i) drew the initialisation, the data order and all device randomness from one shared random stream, so that two arms of the same seed did not share their initial weights or their data order; (ii) moved the TTv2 and c-TTv2 accumulator straight to its symmetry point while charging 62 pulses it did not apply; and (iii) copied the AIHWKIT preset fields into a model with different bounds and no slope spread. The results are kept here unchanged, as preliminary results of that simulator; the revised experiments of Sections 4.2–4.5 replace them. The diagnostic of Appendix F measures how much each repair moves them.

Table 17: Version 1 (superseded). Device catalog (simulator units; Section 3.1). Levels n_{states}, global asymmetry a, saturation γ_+/γ_-, write noise c_{2c} (per pulse, relative to the step), step spread σ_{d2d}, drift time constant τ (training steps), reference error σ_r. Provenance per field is in records/CATALOG.json.

Device	Label	n_{st}	a	γ_+/γ_-	c_{2c}	σ_{d2d}	τ	σ_r
Li-ECRAM (Tang 2018)	PUBLISHED-FIT	1000	0	0.12/0.51	0.3	0.1	none	0.02
MO-ECRAM (Kim 2019)	PUBLISHED-FIT	7089	0	0.42/0.73	2	0.1	none	0.02
Protonic (Onen 2022)	PARTLY-PUBLISHED	1000	0.05	0.20/0.20	0.3	0.1	none	0.02
W/YSZ/WO_x (Chen 2023)	PUBLISHED-FIT	256	0	0.63/0.63	2.048	0.023	none	0.02
Memistor (Widrow 1960, as claimed)	CLAIM+ESTIMATE	1000	0.05	0.10/0.10	0.3	0.2	none	0.02
Memistor (pessimistic)	ESTIMATE	100	0.2	0.30/0.30	1	0.3	1000	0.05
Pencil lead built today	ESTIMATE	100	0.1	0.30/0.30	0.5	0.2	940	0.05

The original headline readings were: plain SGD failed on every published device and task; the Li-ECRAM fit and the protonic device passed or were unresolved on all three tasks with a Tiki-Taka-family algorithm; the original width classifier labelled every arm FLAT (which, as the review noted, is not evidence of equivalence); and the original specification sheet was mostly "not reachable", which led to a post hoc degradation sheet relative to an already imperfect base (a CIFAR base gap of 2.39 pp plus 2 pp admits gaps up to 4.39 pp). The best original analog LM cell was 21.66 % worse in bits per byte than digital Adam (a perplexity ratio of 1.425).

Table 18: Version 1 (superseded). Settings chosen on MNIST validation with the mid device (Appendix B); validation accuracy in %.

Algorithm	Chosen setting	Val. acc.
TTv2	λ 0.1, A lr $\times$10, period 16	97.86
c-TTv2	λ 0.1, A lr $\times$10, period 16, ρ 0.2, reset yes	97.82
AGAD	λ 0.1, A lr $\times$10, period 16, ρ 0.05, β 0.3, reset no	97.18

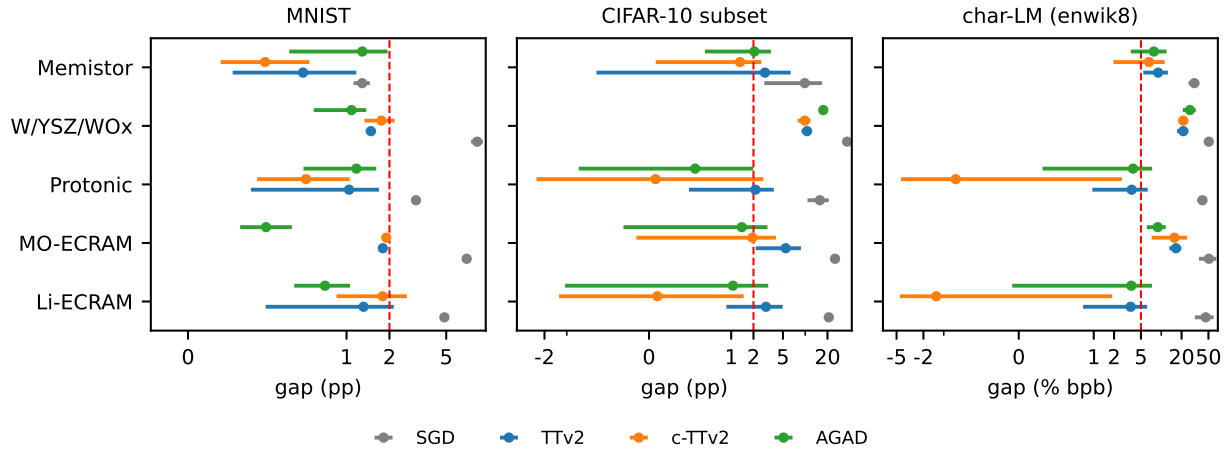


Figure 4: Version 1 (superseded). Original simulator: gap to the exact-update baseline (mean and 95% interval over 5 seeds; symmetric-log axis); the dashed line is the gap line.

Table 19: Version 1 (superseded). Main result (E3): gap to the exact-update baseline, mean and 95% t-interval over 5 seeds, paired by seed. MNIST and CIFAR in percentage points of test accuracy; LM in % of bits per byte. P = the whole interval is below the line (2 pp; 5%), F = wholly above, U = the interval contains the line. Every arm uses the FP validation learning rate.

Device	Algorithm	MNIST gap (pp)	CIFAR-10 subset gap (pp)	char-LM gap (%)
FP baseline (acc. %; bpb)		97.85 [97.79, 97.92]	60.39 [58.64, 62.14]	2.912 [2.822, 3.001]
Li-ECRAM (Tang 2018)	SGD	4.85 [4.73, 4.98] F	20.81 [18.06, 23.57] F	45.58 [31.52, 59.64] F
	TTv2	1.32 [0.49, 2.14] U	2.95 [0.94, 4.96] U	3.52 [0.86, 6.18] U
	c-TTv2	1.79 [0.94, 2.65] U	0.10 [-1.27, 1.47] P	-1.29 [-4.46, 1.87] P
	AGAD	0.86 [0.67, 1.06] P	1.06 [-1.05, 3.17] U	3.60 [-0.09, 7.30] U
MO-ECRAM (Kim 2019)	SGD	6.96 [6.60, 7.31] F	25.29 [22.15, 28.44] F	50.84 [36.24, 65.45] F
	TTv2	1.80 [1.68, 1.91] P	5.45 [2.15, 8.75] F	16.51 [13.17, 19.84] F
	c-TTv2	1.90 [1.81, 1.99] P	1.94 [-0.15, 4.03] U	15.68 [7.21, 24.14] F
	AGAD	0.49 [0.33, 0.66] P	1.39 [-0.31, 3.09] U	8.88 [6.10, 11.65] F
Protonic (Onen 2022)	SGD	3.07 [2.85, 3.29] F	15.77 [10.75, 20.79] F	40.82 [34.31, 47.33] F
	TTv2	1.04 [0.40, 1.69] P	2.13 [0.49, 3.77] U	3.64 [0.98, 6.30] U
	c-TTv2	0.74 [0.44, 1.05] P	0.08 [-2.57, 2.73] U	-0.84 [-4.30, 2.62] P
	AGAD	1.17 [0.73, 1.62] P	0.56 [-0.85, 1.98] P	3.82 [0.32, 7.33] U
W/YSZ/WO_x (Chen 2023)	SGD	8.25 [7.47, 9.02] F	36.82 [33.17, 40.47] F	50.67 [47.53, 53.81] F
	TTv2	1.48 [1.37, 1.60] P	10.59 [8.94, 12.24] F	21.17 [17.11, 25.24] F
	c-TTv2	1.75 [1.33, 2.17] U	9.87 [7.90, 11.84] F	21.23 [17.81, 24.66] F
	AGAD	1.08 [0.79, 1.37] P	17.55 [15.52, 19.58] F	26.59 [20.81, 32.38] F
Memistor (Widrow 1960, as claimed)	SGD	1.29 [1.12, 1.46] P	9.86 [2.80, 16.92] F	30.94 [25.31, 36.57] F
	TTv2	0.73 [0.28, 1.17] P	2.86 [-0.64, 6.35] U	8.99 [5.43, 12.54] F
	c-TTv2	0.49 [0.21, 0.77] P	1.32 [0.08, 2.55] U	6.59 [1.96, 11.23] U
	AGAD	1.29 [0.64, 1.94] P	2.06 [0.68, 3.44] U	7.76 [3.53, 11.99] U

Table 20: Version 1 (superseded). Scale (E4): every arm uses its own validation-chosen learning rate (in parentheses) at every width; gap (pp for CIFAR, % for the LM), mean and 95% interval over 3 seeds paired with FP at its own learning rate. Verdict by the pre-registered rule (Section 4.4).

Task	Width	FP metric (lr)	Li-ECRAM c-TTv2	Li-ECRAM TTv2	Protonic c-TTv2	Protonic TTv2
CIFAR	16	55.70 (0.0003)	0.64 [-1.53, 2.82] (0.003)	1.86 [-0.64, 4.35] (0.001)	-2.13 [-7.14, 2.88] (0.003)	0.41 [-1.44, 2.26] (0.001)
CIFAR	32	60.15 (0.001)	-0.40 [-3.13, 2.33] (0.001)	2.12 [-1.66, 5.90] (0.001)	-1.88 [-6.35, 2.59] (0.003)	-0.21 [-5.27, 4.84] (0.003)
CIFAR	64	60.87 (0.0003)	-1.34 [-5.57, 2.89] (0.001)	1.11 [0.17, 2.05] (0.001)	-1.30 [-3.56, 0.96] (0.003)	0.33 [-2.43, 3.09] (0.001)
		verdict:	FLAT	FLAT	FLAT	FLAT
LM	64	3.193 (0.0001)	-0.55 [-8.58, 7.48] (0.0001)	-0.31 [-6.26, 5.63] (0.0003)	-7.21 [-11.77, -2.65] (0.0003)	-3.45 [-6.44, -0.46] (0.0003)
LM	128	2.881 (0.0001)	-0.09 [-7.23, 7.04] (0.0001)	-0.60 [-8.73, 7.53] (0.0003)	-6.38 [-12.53, -0.23] (0.0003)	-3.88 [-10.90, 3.14] (0.0003)
LM	256	3.100 (3e-05)	-13.73 [-14.66, -12.79] (0.0001)	-9.01 [-10.21, -7.82] (0.0001)	-12.97 [-13.57, -12.36] (0.0001)	-8.48 [-9.67, -7.29] (0.0001)
		verdict:	FLAT	SHRINKS	FLAT	SHRINKS

Table 21: Version 1 (superseded). Device specification sheet (E5) for TTv2: the weakest grid value of each parameter that keeps the gap under the line (2 pp CIFAR; 5% LM) with every other parameter at the protonic base (levels 1000, a 0.05, γ 0.2, c_{2c} 0.3, no drift, σ_r 0.02); in parentheses the interpolated crossing. †: the base itself fails and the value shown is the improvement needed; *: passes over the whole grid; n/r: not reachable on the grid. Last column: the joint specification (every axis at its threshold at once), mean gap over 3 seeds, round 1 or the tightened round 2.

Size	Levels	Asym. a	Nonlin. γ	Noise c_{2c}	Retention τ	Ref. err. σ_r	Joint spec
CIFAR 16	n/r	n/r	n/r	n/r	n/r	$\leq$ 0 (0.0104)†	–
CIFAR 32	$\geq$ 2000 (1488)†	n/r	n/r	$\leq$ 0 (0.17)†	n/r	$\leq$ 0 (0.0179)†	–
CIFAR 64	n/r	n/r	n/r	n/r	n/r	n/r	–
LM 64	$\geq$ 500 (292)	$\leq$ 0.2 (0.231)	$\leq$ 0.2 (0.379)	$\leq$ 1 (1.12)	$\geq$ 1e4 (7e3)	$\leq$ 0.02 (0.0239)	r1: 4.74 pass
LM 128	$\geq$ 1000 (612)	$\leq$ 0.2 (0.21)	$\leq$ 0.2 (0.281)	$\leq$ 0.3 (0.727)	$\geq$ 1e4 (8e3)	$\leq$ 0.02 (0.021)	r1: 2.45 pass
LM 256	n/r	n/r	n/r	n/r	n/r	n/r	–

Table 22: Version 1 (superseded). Degradation thresholds (descriptive, added after the one-at-a-time data were seen; DEVIATIONS-PAPER.md D1): the weakest grid value of each parameter whose gap stays within the line (2 pp CIFAR; 5% LM) of the protonic base device's own gap (column 3), all other parameters at the base; interpolated crossing in parentheses; * the whole grid stays within; n/r the nearest grid step already leaves the band.

Alg.	Size	Base gap	Levels	Asym. a	Nonlin. γ	Noise c_{2c}	Retention τ	Ref. err. σ_r
SGD	CIFAR 32	19.80	$\geq$ 1000 (520)	$\leq$ 0.05 (0.0813)	$\leq$ 0.2 (0.266)	$\leq$ 1 (1.3)	$\geq$ 1e4 (7e3)	$\leq$ 0.1 (0.15)
SGD	LM 128	39.85	$\geq$ 200 (143)	$\leq$ 0.4*	$\leq$ 0.2 (0.423)	$\leq$ 4*	$\geq$ 1e4 (3e3)	$\leq$ 0.5*
TTv2	CIFAR 16	4.03	$\geq$ 100*	$\leq$ 0.2 (0.204)	$\leq$ 0.5 (0.606)	$\leq$ 0.3 (0.789)	$\geq$ 1e5 (7e4)	$\leq$ 0.02 (0.0228)
TTv2	CIFAR 32	2.39	$\geq$ 500 (265)	$\leq$ 0.2 (0.209)	$\leq$ 0.5 (0.578)	$\leq$ 2 (2.31)	$\geq$ 1e5 (2e4)	$\leq$ 0.02 (0.0241)
TTv2	CIFAR 64	7.32	$\geq$ 100*	$\leq$ 0.2 (0.203)	$\leq$ 2*	$\leq$ 2 (2.62)	$\geq$ 1e5 (4e4)	$\leq$ 0.02 (0.0244)
TTv2	LM 64	3.18	$\geq$ 200 (130)	$\leq$ 0.2 (0.284)	$\leq$ 0.5 (0.717)	$\leq$ 2 (2.03)	$\geq$ 1e4 (5e3)	$\leq$ 0.02 (0.0326)
TTv2	LM 128	4.25	$\geq$ 200 (167)	$\leq$ 0.2 (0.276)	$\leq$ 0.5 (0.705)	$\leq$ 1 (1.67)	$\geq$ 1e4 (6e3)	$\leq$ 0.02 (0.028)
TTv2	LM 256	16.29	$\geq$ 100*	$\leq$ 0.4*	$\leq$ 2*	$\leq$ 2 (2.64)	$\geq$ 1e4 (4e3)	$\leq$ 0.02 (0.0351)

Table 23: Version 1 (superseded). Pairwise sensitivity (E5): the three largest interactions per task, interaction = gap(a,b) − [gap(a) + gap(b) − gap(base)].

Task	Pair	Gap(a,b)	Singles	Interaction
CIFAR	asym=0.1, tau_steps=1e3	46.64	1.82 + 33.96	13.25
CIFAR	asym=0.2, tau_steps=1e3	40.24	1.92 + 33.96	6.75
CIFAR	gamma=0.5, tau_steps=1e3	41.17	3.55 + 33.96	6.05
LM	c2c=1, tau_steps=1e3	76.94	5.27 + 33.74	42.18
LM	gamma=0.5, tau_steps=1e3	76.63	6.28 + 33.74	40.85
LM	asym=0.1, tau_steps=1e3	73.70	4.09 + 33.74	40.11

Table 24: Version 1 (superseded). System energy per weight per row on the character LM (E6), in pJ: forward and backward MVMs at the measured 12.4 TOPS/W of [37], device pulses (simulated count × published or estimated energy per pulse), pulse lines and generation, and the TT-family reads. Ratio = H100 at its measured utilisation over the analog total; the range sets every input at its low or high end (records/ENERGY-SYSTEM.json).

Device	Algorithm	Pulses/w/row	MVM	Device	Lines	TT	Ratio	Range
Li-ECRAM (Tang 2018)	AGAD	0.034	0.323	0.0000	0.0013	0.00003	32.4	16.2–234
Li-ECRAM (Tang 2018)	c-TTv2	0.042	0.323	0.0000	0.0013	0.00003	32.3	16.2–234
Li-ECRAM (Tang 2018)	SGD	0.001	0.323	0.0000	0.0012	0.00000	32.4	16.2–234
Li-ECRAM (Tang 2018)	TTv2	0.038	0.323	0.0000	0.0013	0.00003	32.4	16.2–234
MO-ECRAM (Kim 2019)	AGAD	0.267	0.323	0.0003	0.0015	0.00003	32.3	16.2–232
MO-ECRAM (Kim 2019)	c-TTv2	0.339	0.323	0.0003	0.0015	0.00003	32.3	16.2–232
MO-ECRAM (Kim 2019)	SGD	0.007	0.323	0.0000	0.0013	0.00000	32.4	16.2–234
MO-ECRAM (Kim 2019)	TTv2	0.300	0.323	0.0003	0.0015	0.00003	32.3	16.2–232
Protonic (Onen 2022)	AGAD	0.026	0.323	0.0001	0.0038	0.00003	32.1	15.8–228
Protonic (Onen 2022)	c-TTv2	0.035	0.323	0.0001	0.0038	0.00003	32.1	15.8–228
Protonic (Onen 2022)	SGD	0.001	0.323	0.0000	0.0018	0.00000	32.3	16.1–233
Protonic (Onen 2022)	TTv2	0.033	0.323	0.0001	0.0037	0.00003	32.1	15.8–228
Memistor (Widrow 1960, as claimed)	AGAD	0.017	0.323	262122.1163	0.0012	0.00003	4e-05	2.69e-05–4.26e-05
Memistor (Widrow 1960, as claimed)	c-TTv2	0.022	0.323	323881.6451	0.0012	0.00003	3.24e-05	2.18e-05–3.45e-05
Memistor (Widrow 1960, as claimed)	SGD	0.002	0.323	30015.2982	0.0012	0.00000	0.000349	0.000235–0.000372
Memistor (Widrow 1960, as claimed)	TTv2	0.022	0.323	326250.3670	0.0012	0.00003	3.21e-05	2.16e-05–3.42e-05
W/YSZ/WO_x (Chen 2023)	AGAD	0.007	0.323	0.0000	0.0015	0.00003	32.3	16.2–233
W/YSZ/WO_x (Chen 2023)	c-TTv2	0.010	0.323	0.0000	0.0015	0.00003	32.3	16.2–233
W/YSZ/WO_x (Chen 2023)	SGD	0.000	0.323	0.0000	0.0013	0.00000	32.4	16.2–234
W/YSZ/WO_x (Chen 2023)	TTv2	0.009	0.323	0.0000	0.0015	0.00003	32.3	16.2–233

F Revision history and response to the independent review

History. Version 1 (October 25, 2026) reported the experiments of Appendix E. An independent technical review (October 26, 2026) recommended major revision with thirteen findings. Version 2 answered them; the full response, finding by finding, with the checks that confirmed or qualified each one, is `RESPONSE-TO-REVIEW.md` in the release. The revision plan was committed before any revision run (`PLAN-REVISION.md`); its deviations R1–R9 are recorded in `DEVIATIONS-PAPER.md`. Version 3 (October 26, 2026) answered a second review (eight findings; `RESPONSE-TO-REVIEW-2.md`, plan `PLAN-REVISION-2.md`): the pulse-fidelity experiments of Section 4.3, the recentred fixed-symmetry control, the width contrasts paired by seed and the revised settings tables. Version 4 (October 27, 2026) answers a third review (five findings; `RESPONSE-TO-REVIEW-3.md`, plan `PLAN-REVISION-3.md`, committed before its runs): the per-sample stream is compared with the literal one paired by draw, with bootstrap bounds, in a desk rerun of the update probes that keeps the draws; the cross-cell dependence is measured by per-pair correlations that cannot cancel; the training comparison reports both arms' labels and every rank reversal; the open training rerun of the one material cell was done; and an interval that includes the width margin is no longer read as growth below it. Version 5 (October 27, 2026) answers a fourth review (three findings; `RESPONSE-TO-REVIEW-4.md`, plan `PLAN-REVISION-4.md`, committed before its runs). Version 4's five-seed rerun of the material SGD cell had joined, by seed only, two stream runs whose initialisation and example order differed from their controls' (the random draws depend on the GPU model), and it had put a seed-0 difference down to GPU rounding; its figures (stream gap 11.04 pp, change -0.47 pp) are withdrawn as an unmatched, mixed-hardware diagnostic. The cell was rerun as matched triples on one GPU, and the analysis now refuses a seed whose arms do not share both hashes. The saved pulse draws were added to the deposit inventory with a command that rebuilds the probe tables from them, the probe table was split in two for legibility, and the abstract states that its probe count is a point-estimate count. Version 6 (October 27, 2026) answers a fifth review (one finding and two wording corrections; `RESPONSE-TO-REVIEW-5.md`; no experiment was run and no number changed). The HTML edition of versions 4 and 5 was incomplete: it lacked Sections 5–9, the contribution and acknowledgment statements and two appendices, it printed the rows of the simulator-checks table under the sensitivity table's caption, and nine references were unresolved. The converter had read two adjacent tables with different column-width units as one and dropped everything between them. The conversion now rewrites paragraph-width columns, and a new check refuses any HTML that lacks a heading or a table's own rows or that holds an unresolved reference. The two wording corrections are in Section 4.3 and Section 9. Version 7 (October 27, 2026) answers a sixth review (one cosmetic finding; `RESPONSE-TO-REVIEW-6.md`; no experiment was run, and no number or sentence of the PDF changed except this history). The HTML edition of version 6 printed the rule arguments as text before "Seed" in the header of Table 9. The conversion now removes the table-rule commands, and the HTML check refuses any leftover rule argument or raw command. Version 8 (October 27, 2026) answers a seventh review, which found no actionable defect (`RESPONSE-TO-REVIEW-7.md`; no experiment was run and no number changed). It carries two changes ordered by the author: his official name, Anthony DeMarco, replaces the name printed in versions 1 to 7, and the contribution statement credits ChatGPT's reviews. Version 9 (October 28, 2026) corrects the deviation count R1–R6 to R1–R8, found by the deposit's clean-room regeneration. Version 10 (October 28, 2026) gives the public archive, its DOI and its GitHub mirror in Section 9; no number changed. Version 11 (October 28, 2026) answers an eighth review, a code audit and reproduction from the public archive (`RESPONSE-TO-REVIEW-8.md`; no experiment was run). Figure 4 (version 1, superseded) drew two negative LM means at +0.05, with their intervals shifted, and its legend overlapped an axis label; both are corrected. The energy model now counts a classification run's optimizer steps

from the rows its layers trained: the run records' sample count includes each epoch's incomplete last batch, which training skips, so the transfer reads were overstated by 0.16 %. No printed number changed; the largest change in a total energy is 0.0005 % in the current projection and 0.0014 % in version 1's. Section 3 now discloses an off-by-one in the LM sampler. The simulator's ideal reset of a partial column selection under AIHWKIT bounds, which no result used, no longer fails, and a new check covers it. Version 12 (October 29, 2026) answers a ninth review, a second audit and reproduction from the public archive (`RESPONSE-TO-REVIEW-9.md`; no experiment was run). The width study's Welch interval now takes the t quantile at the Welch–Satterthwaite degrees of freedom; versions 2 to 11 rounded them down, which the plan does not state (deviation R9). This changes one verdict, protonic-inspired c-TTv2 on CIFAR, from inconclusive to grows (Section 4.4), and moves the other Welch intervals by at most 0.18 in their units; the paired contrast is unchanged. The width section now says that across widths a seed shares its example order, not its initialisation; the excess correlation of Table 8 is defined as an excess in quadrature; and Table 16 labels its version-1 values as superseded. Version 13 (October 31, 2026) answers a tenth review, of version 12 and version 1.2 of the archive (`RESPONSE-TO-REVIEW-10.md`; no experiment was run and no number changed). It found the Welch correction right and four small slips outside the results, all corrected: the archive's README said that no other printed number changed, although all eight Welch intervals moved, and counted eight reviews; the caption of Table 11 dated the rounded degrees of freedom to versions 1 to 11, not 2 to 11 (version 1 used another width test); and a self-test of the name check left a PDF open, so its clean-up failed on Windows. Version 14 (October 31, 2026) answers an eleventh review, of version 13 (`RESPONSE-TO-REVIEW-11.md`; no experiment was run and no result changed). It cites Byun et al. [16] and states what this study adds to it; prints the GPU energy equation with both input ranges (the low end, 7.05 pJ, is 500 W at 43 %; version 13 named only the utilisation range); says which digital costs of the TT family are inside the energy ratio and that the ratio is a per-operation, read-dominated figure; calls the sensitivity sheet an exploratory device sensitivity analysis, in the subtitle too, and states that one-at-a-time ranges do not bound joint deviations; adds Figure 1; adds a control ladder to the limitations and computational follow-ups to the roadmap; and moves correction history from Sections 3, 4.3, 4.4 and 4.5 to this appendix. Version 15 (November 1, 2026; this paper) answers a twelfth review, of version 14 and version 1.4 of the archive (`RESPONSE-TO-REVIEW-12.md`; no experiment was run and no result changed), which recomputed the statistics and the energy accounting from the records and ran the archive's fast reproduction. Figure 1 and its caption now show the selected settings (TTv2 and c-TTv2 reset A with pulses, AGAD does not; TTv2 has no chopper), and its check takes them from the main experiment's settings and records; Section 2 cites Agarwal et al. before the Tiki-Taka studies; the ordinary and paired intervals take the exact t quantile, as the Welch interval already did, which moves 16 endpoints in Tables 4 and 5 by 0.01 and changes no classification; Section 5 states that the reset's extra reads carry a buffer allowance; and outside its reference environment the archive's reproduction compares its one CPU re-simulation of float32 pulse sequences (the AIHWKIT comparison) under a stated numeric tolerance.

What was repaired.

1. **Random streams.** The initialisation, the data order, each tile's device population and the pulse/noise draws now have separate generators; every record carries the hashes of its logical initialisation and data order, and every arm of a seed shares them (checked). The old single stream is kept as a mode that replays the original simulator bit for bit.

2. **The reset.** The original reset moved A to its symmetry point and charged 62 pulses it never applied; applied through the device model those pulses leave 97.55 % of the displacement

(from $u = 0.8$ the state only reaches 0.7835, against a symmetry point of 0.125; about 11,513 alternating pulses would be needed for a hundredfold reduction). The repaired pulse reset applies and counts every pulse; the ideal reset is reported only as a labelled variant.

3. **The AIHWKIT presets** are mapped one to one (shifted bounds, bound-scaled slopes, slope spread), and their pulse responses were compared with AIHWKIT's own (Section 4.1): the mapping is accepted.

4. **Measured, not assumed:** the clipping of the pulse probabilities and the lost update are recorded per layer.

The 8 checks of the repairs each fail on the old behaviour or a planted fault (8 faults; ALL_OK on the desk and on every pod).

How much the repairs moved the original conclusions (diagnostic D1). Table 25 runs the original and the repaired simulator side by side at the original selections, 3 seeds, on MNIST and a 10-epoch CIFAR slice (234 runs).

- **The random streams** moved individual cells by up to 1.68 pp on MNIST and 3.89 pp on CIFAR (material, i.e. more than 1 pp or a label change, in 4 of 8 and 7 of 8 cells). With three seeds this is within the noise: the repaired FP-SGD alone spans 52.0–57.2 % on the CIFAR slice. The matched controls did not reduce the paired-gap spread here (median 1.78 against 0.76 pp).

- **The reset** moved cells by up to 1.38 pp on MNIST and 4.51 pp on CIFAR. The ideal reset gave no hidden advantage: a physical reset (pulse, alternating pairs or none) had an equal or smaller gap in 19 of 24 cells.

- **The mapping** moved Li-ECRAM by at most 0.52 pp on MNIST and 1.08 pp on CIFAR.

- **Unchanged:** plain SGD fails by far (MNIST 3.09–6.99 pp, CIFAR slice 10.35–17.87 pp).

Wording withdrawn or narrowed. "Near-digital" is not used. "Plain SGD fails on every device" became "every plain-SGD arm at the FP-selected learning rate exceeded the margin". "The gap does not grow with width" became the equivalence statement of Section 4.4. "Two device properties decide" and "a few hundred levels suffice" became conditional statements about the sweeps of Section 4.5. "The memistor lacked density, speed and energy, not accuracy" became "an optimistic historical-device scenario attained useful accuracy in this simulator". "32× today with the periphery counted" became the conditional projection of Section 5. "A bench replication" became "a proposed, unbuilt bench replication protocol". "Every matrix is analog" became "the learned weight matrices use the device model; attention products and other operations are digital".

Version 3's update probes (superseded). Table 26 is version 3's analysis of the update probes (pod sitting r6). Its sampling floor for the per-sample stream added the two methods' variances as if their draws were independent, although the per-sample stream replays the literal stream's line firings, so the floor was too large and every excess of the per-sample stream came out as zero. Version 4 replaces it by the paired analysis of Section 4.3 (Table 7). The surrogate's verdicts did not depend on this floor. The statistic of version 3, a pooled, signed normalised covariance, stays within 0.020 in absolute value on the same draws, because positive and negative terms cancel and unequal cell variances are pooled.

Table 25: Diagnostic D1: mean gap to FP-SGD of the same simulator (pp, 3 seeds; P/F/U against 2 pp) at the original selections. Columns: orig. = the original simulator; ideal = the repaired random streams with the original ideal reset; then the reset replaced by the pulse reset (pulse), 31 alternating pulse pairs (alt.), or none; publ. = the published-style TTv2. The AIHWKIT-mapped Li-ECRAM row uses the pulse reset for TTv2 and c-TTv2. CIFAR here is a 10-epoch slice.

Task	Device	Alg.	orig.	ideal	pulse	alt.	none	publ.
MNIST	Protonic-insp.	SGD	3.09F	4.58U				
MNIST	Protonic-insp.	TTv2	1.04U	1.60U	0.23P	0.66P	1.02P	0.99P
MNIST	Protonic-insp.	c-TTv2	0.81P	0.62U	0.44P	0.97U	1.41P	
MNIST	Protonic-insp.	AGAD	1.21P	0.99P				
MNIST	Li (old map)	SGD	4.79F	6.47U				
MNIST	Li (old map)	TTv2	1.33U	1.19U	0.39P	0.36P	0.66P	1.43P
MNIST	Li (old map)	c-TTv2	1.51U	0.98P	0.80P	0.68P	1.00P	
MNIST	Li (old map)	AGAD	0.96P	1.00P				
MNIST	Li (AIHWKIT)	SGD		6.99U				
MNIST	Li (AIHWKIT)	TTv2			0.40P			
MNIST	Li (AIHWKIT)	c-TTv2			0.70P			
MNIST	Li (AIHWKIT)	AGAD		0.57P				
CIFAR-10ep	Protonic-insp.	SGD	14.24F	10.35F				
CIFAR-10ep	Protonic-insp.	TTv2	5.28U	2.53U	2.04U	-0.83U	-1.97U	4.17U
CIFAR-10ep	Protonic-insp.	c-TTv2	4.59U	3.41U	5.97U	-0.43U	0.65U	
CIFAR-10ep	Protonic-insp.	AGAD	-0.70P	-2.94U				
CIFAR-10ep	Li (old map)	SGD	17.69F	16.91F				
CIFAR-10ep	Li (old map)	TTv2	4.35F	2.51U	2.98U	0.57U	0.78U	5.41U
CIFAR-10ep	Li (old map)	c-TTv2	6.08U	2.27U	1.63U	1.48U	0.99U	
CIFAR-10ep	Li (old map)	AGAD	0.39U	-1.34U				
CIFAR-10ep	Li (AIHWKIT)	SGD		17.87F				
CIFAR-10ep	Li (AIHWKIT)	TTv2			2.37U			
CIFAR-10ep	Li (AIHWKIT)	c-TTv2			2.71U			
CIFAR-10ep	Li (AIHWKIT)	AGAD		-1.02U				

Random realisations of the probe and rerun states. The probes draw from a generator of their own. On the pod (version 3) the MNIST and CIFAR probe runs reproduced their main-table records exactly; on the desk (Quadro P2200) 0 of 3 do, and none shares its initialisation with its version-3 run (the MNIST run not its example order either): the simulator's generators run on the GPU, and the CUDA random kernels assign draws to threads by the GPU's multiprocessor count. The desk's probe runs are therefore fresh training realisations, and the probed states are not version 3's; the comparison within each probe is unaffected, since all three updates start from the same saved state. LM training is not bitwise reproducible even on one card type: configurations repeated exactly in the records differ by up to 0.31 % in bits per byte (CIFAR 0.00 %). The material SGD cell of Section 4.3 was rerun as matched triples on the desk. The pod's runs of seeds 0–2 are other random realisations of the same seeds, not a rounding variant of them: the simulator's generators run on the GPU, and the CUDA random kernels assign draws to threads by the GPU's multiprocessor count, so another GPU model draws another initialisation and example order (as predicted, the hashes differ from both the pod's and the earlier desk runs'). Per seed the two realisations' gaps differ by up to 9.97 pp, and at the pod's three matched seeds the change was -1.42 pp [-4.84, 2.00]: its gap depends strongly on the realisation.

The held-symmetry anchor. With the symmetry point held at mid-range, version 2 borrowed the ordinary base (whose range is $[-1.125, +0.875]$) as this sweep's anchor and omitted its zero point (deviation R5); version 3 ran both with each device's range re-centred (Section 4.5).

Table 26: Version 3 (superseded: its sampling floor for seq treated the paired draws as independent). Update fidelity (experiment A1; Li = Li-ECRAM, MO = MO-ECRAM, Prot. = protonic-inspired): one training batch applied repeatedly from the same state by the aggregate surrogate (sur.), the per-sample stream (seq) and the literal pulse-by-pulse stream (slot). Lost = unsigned requested pulse mass removed by the probability clipping (%); g = the slot stream's realised gain along the requested update; then surrogate / seq against slot: gain difference (%), excess relative mean difference beyond the sampling floor, median per-cell variance ratio; faithful by the pre-registered lines (gain within 10 %, excess ≤ 0.10, variance ratio in [0.5, 2]; for seq 5 %, 0.05, [0.8, 1.25]).

Case	Layer	Upd.	Lost	g	Δg	excess	var. ratio	faithful
CIFAR MO AGAD	3 (128×1024)	600	32.1	0.431	0.7 / -0.2	0.07 / 0.00	1.08 / 1.00	yes / yes
CIFAR MO AGAD	4 (10×128)	600	92.1	0.027	1.8 / 0.3	0.18 / 0.00	1.20 / 1.02	no / yes
CIFAR MO AGAD	3 (128×1024)	2000	15.0	0.700	1.3 / -0.0	0.03 / 0.00	1.07 / 1.00	yes / yes
CIFAR MO AGAD	4 (10×128)	2000	96.5	0.004	32.7 / 0.6	0.37 / 0.00	1.22 / 1.00	no / yes
LM Prot. c-TTv2	0 (128×256)	750	7.6	0.717	0.3 / 0.1	0.01 / 0.00	1.00 / 1.00	yes / yes
LM Prot. c-TTv2	9 (256×128)	750	0.0	0.972	-1.4 / -0.5	0.00 / 0.00	0.71 / 1.01	yes / yes
LM Prot. c-TTv2	0 (128×256)	2250	32.5	0.547	-0.1 / 0.0	0.04 / 0.00	1.02 / 1.00	yes / yes
LM Prot. c-TTv2	9 (256×128)	2250	0.0	0.882	0.2 / 0.3	0.00 / 0.00	0.73 / 1.01	yes / yes
MNIST Li c-TTv2	0 (256×784)	1000	21.6	0.759	0.1 / -0.0	0.03 / 0.00	1.05 / 1.00	yes / yes
MNIST Li c-TTv2	2 (10×128)	1000	3.7	0.990	-0.0 / -0.0	0.00 / 0.00	1.26 / 0.98	yes / yes
MNIST Li c-TTv2	0 (256×784)	3500	25.7	0.696	-0.3 / -0.0	0.01 / 0.00	1.09 / 1.00	yes / yes
MNIST Li c-TTv2	2 (10×128)	3500	5.0	0.922	-1.1 / 0.1	0.00 / 0.00	1.55 / 1.00	yes / yes